

Cerebellar Purkinje Cells Control Posture in Larval Zebrafish (*Danio rerio*)

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Franziska Auer, Katherine Nardone, Koji Matsuda, Masahiko Hibi, David Schoppik 

Depts. of Otolaryngology, Neuroscience & Physiology, and the Neuroscience Institute, NYU Grossman School of Medicine, New York, United States • Division of Biological Science, Graduate School of Science, Nagoya University, Nagoya, Japan

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eLife Assessment

This **important** study successfully applies an innovative chemogenetic tool to investigate cerebellar function to advance our understanding of the contributions of Purkinje cell populations to postural control in larval zebrafish. The evidence supporting the conclusions is **convincing** and supported by rigorous statistical analysis. The study highlights the power of combining genetically targeted perturbations with quantitative high-throughput behavioral analysis and original microscopy tools.

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Abstract

Cerebellar dysfunction leads to postural instability. Recent work in freely moving rodents has transformed investigations of cerebellar contributions to posture. However, the combined complexity of terrestrial locomotion and the rodent cerebellum motivate new approaches to perturb cerebellar function in simpler vertebrates. Here, we adapted a validated chemogenetic tool (TRPV1/capsaicin) to describe the role of Purkinje cells — the output neurons of the cerebellar cortex — as larval zebrafish swam freely in depth. We achieved both bidirectional control (activation and ablation) of Purkinje cells while performing quantitative high-throughput assessment of posture and locomotion. Activation modified postural control in the pitch (nose-up/nose-down) axis. Similarly, ablations disrupted pitch-axis posture and fin-body coordination responsible for climbs. Postural disruption was more widespread in older larvae, offering a window into emergent roles for the developing cerebellum in the control of posture. Finally, we found that activity in Purkinje cells could individually and collectively encode tilt direction, a key feature of postural control neurons. Our findings delineate an expected role for the cerebellum in postural control and vestibular sensation in larval zebrafish, establishing the validity of TRPV1/capsaicin-mediated perturbations in a simple, genetically-tractable vertebrate. Moreover, by comparing the contributions of Purkinje cell ablations to posture in time, we uncover signatures of emerging cerebellar control of posture across early development. This work takes a major step towards understanding an ancestral role of the cerebellum in regulating postural maturation.

Introduction

In vertebrates, cerebellar activity underlies proper posture, defined as the relative orientation of body parts in space ^{1–8}. The cerebellum integrates sensory information from vestibular (balance), visual, and proprioceptive systems ³. These sensations are transformed into precise and coordinated adjustments in muscle tone and contraction allowing animals to control posture ⁹. Disruptions to mature cerebellar function lead to instability, unsteady gait and a compromised sense of balance ¹⁰. Development of the cerebellum coincides with postural maturation, and early development of the cerebellum has been extensively studied ^{11–13}. Notably, changes to morphology and activity of the output neurons of the cerebellar cortex, Purkinje cells, are thought to underlie the gradual refinement of motor control ¹⁴. To date, the contributions of developing Purkinje cells to postural control remains poorly understood.

Kinematic quantification by pose estimation in rodents ^{15,16} has opened a window into cerebellar contributions to postural behaviors in health and disease ^{8,17,18}. However, terrestrial gait and locomotion are complex and especially difficult to study during development; tracking and analysis is often limited to measures such as the time of head elevation or the duration of walking stance ¹⁹. In contrast, the biophysical challenges of maintaining posture underwater are straightforward to define ^{20–22}. For example, larval zebrafish balance in the pitch axis (nose-up/nose-down) by timing locomotion to countermand gravity-induced destabilization ^{23,24} and by coordinated use of paired appendages (fins) and axial musculature (trunk) ²⁵. The small size and rapid development of the larval zebrafish allow high-throughput measurements of postural control (i.e. pitch-axis kinematics and fin/trunk coordination) from freely swimming subjects ²⁶.

The larval zebrafish is a powerful model to investigate cerebellar development and function ²⁷. Anatomically, the zebrafish cerebellum shares the same circuit structure as the mammalian cerebellum ²⁸. The zebrafish cerebellum is compartmentalized into regions with distinct response properties and output targets ^{28–32}. Multimodal representations were found in both cerebellar granule cells ^{33,34} and Purkinje cells ^{31,35}. Functional assays established a role for the larval zebrafish cerebellum in motor control, sensorimotor integration and predictive neural processing, particularly in response to visual input ^{36–44}. Finally, brain-wide imaging studies have established balance-relevant sensitivity in the cerebellum, identifying neurons that encode body angle and velocity ⁴⁵ and neurons responsive to direct inner-ear stimulation ⁴⁶. Overwhelmingly, this work has been done in reduced or restrained preparations, limiting insight into the cerebellar contribution to natural behaviors.

Powerful new opto- and chemogenetic ⁴⁷ approaches allow control of particular cerebellar cell types, reviewed in ⁴⁸. Recent work used such activation/inhibition to investigate cerebellar contributions to sensorimotor ^{49–52} and non-sensorimotor behaviors ^{53–56} in health and disease ^{57–59}. Both approaches come with technical hurdles: optogenetics requires targeting light to the cerebellum, a particular challenge when untethered animals can move freely in depth, while chemogenetics uses bioactive co-factors ⁶⁰. A chemogenetic approach to cerebellar control with a non-bioactive ligand would be a welcome advance, particularly to study posture without visual interference (i.e. in the dark). One validated path forward is to express the rat non-selective cation channel TRPV1 and its ligand capsaicin in zebrafish ⁶¹. The endogenous zebrafish TRPV1 channel is capsaicin-insensitive ⁶², so targeted expression of rat TRPV1 allows cell-type specific control: low-doses of capsaicin can activate sensory and hypothalamic neurons while high-doses are excitotoxic ⁶¹. Capsaicin can be dissolved in water and is readily absorbed by freely-swimming larval zebrafish, sidestepping invasive procedures and the need for visible light. Finally, the conductance of a TRP channel is ~1000x that of a channelrhodopsin ⁶³ suggesting that even low levels of TRPV1 expression will be biologically effective.

Here we used the TRPV1/capsaicin system to investigate the contribution of cerebellar Purkinje cells to postural behaviors as larval zebrafish swam freely in depth. Both activation and ablation of Purkinje cells could induce changes in pitch axis posture. Ablation in older larvae resulted in bigger disruptions to posture, allowing inference of the functional consequences of cerebellar development. Furthermore, ablation of Purkinje cells in older larvae disrupted the coordination of trunk and paired appendages (fins), impairing vertical navigation. Finally, we could reliably decode pitch-tilt direction from patterns of Purkinje cell activity. Taken together our results establish a clear role for the cerebellum in larval zebrafish postural control, even during the earliest stages of development. More broadly, our work establishes a new method to manipulate cerebellar output while performing quantitative high-throughput measures of unconstrained posture and locomotion. Our data are therefore a step towards defining an ancestral role for the highly-conserved cerebellum in postural control.

Results

A new tool for chemogenetic activation or ablation of Purkinje cells

We used a new tool to control Purkinje cells: the transgenic line *Tg(aldoca:TRPV1-tagRFP)*. Fish in this line express rat TRPV1, a capsaicin-sensitive non-selective cation channel, exclusively in all cerebellar Purkinje cells (**Figures 1A** and **1B**).

Endogenous zebrafish TRPV1 channels are insensitive to capsaicin. Previous descriptions of rat TRPV1 in zebrafish sensory and hypothalamic neurons establish dose-dependent chemogenetic manipulation. We expect low-doses of capsaicin to depolarize Purkinje cells (**Figure 1C**, left), while high-doses should be excitotoxic (**Figure 1C**, right).

First, we assayed capsaicin concentrations and incubation times to identify a dose that would achieve long-term depolarization without cell death. We co-expressed a nuclear-targeted calcium indicator, GCaMP6f (**Figure 1D**) in all neurons using the *Tg(elavl3:h2B-GCaMP6f)* line for longitudinal imaging of neuronal activity. Previous work used 1 μ M of capsaicin for long-term activation. We therefore imaged the cerebellum of *Tg(aldoca:TRPV1-tagRFP);Tg(elavl3:h2B-GCaMP6f)* fish prior to and 3, 6, and 9 hours after 1 μ M capsaicin treatment (**Figure 1D**). We screened fish for comparable brightness and selected fish that had clearly visible expression but were not overly bright (Methods).

Prolonged exposure to a low dose of capsaicin increased cerebellar activity (**Figures S1A** and **1E**). At each timepoint, TRPV1-expressing cells showed increased intensity relative to a pre-capsaicin baseline, while TRPV1-negative cells did not (**Figure 1E**, 3/6 hours post 1 μ M capsaicin: 28%/20%/TRPV1+ cells $F/F_0 > 2$; 40 cells from 3 fish vs. 0%/0%/TRPV1-cells $F/F_0 > 2$; 44 cells from 4 fish; activated cells after 6 h of capsaicin treatment: 0/44 TRPV1-vs. 8/40 TRPV1+; Fisher's exact test: $p = 0.0018$).

Different cells showed increased activity at the 3, 6, and 9 hour timepoints, and the same cells were differentially active at different timepoints. We interpret this as evidence that 1 μ M of capsaicin could sporadically activate subsets of Purkinje cells. Notably, in one fish that had particularly strong tagRFP expression we observed a small number of neurons at the 9 h timepoint with bright, speckled fluorescence suggestive of cell death (**Figure S1C**). We did not observe any signs of cell death at the 6 h timepoints (6 TRPV1+ fish at 6 h post 1 μ M capsaicin). We therefore set an upper limit of 6 h of exposure to 1 μ M capsaicin for activation experiments.

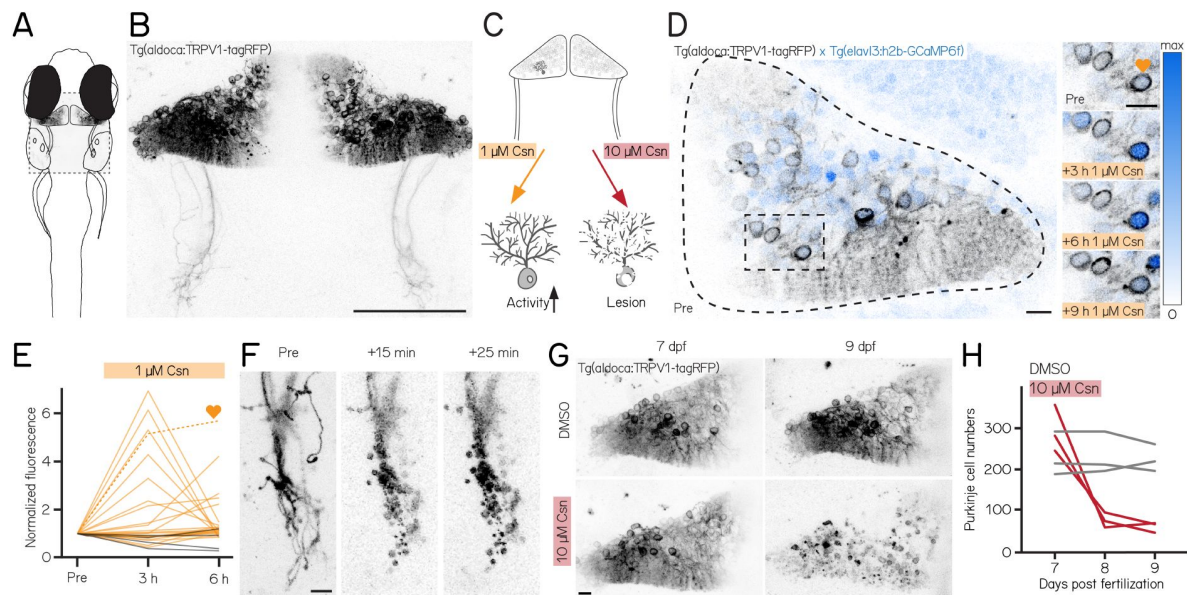


Figure 1

A chemogenetic approach allows dose-dependent activation and lesion of Purkinje cells in the cerebellum.

(A) Schematic of a larval zebrafish overlaid with a confocal image of labeled Purkinje cells in the cerebellum. Gray rectangle corresponds to field of view in (B). (B) Confocal image of Purkinje cells in the cerebellum of a 7 days post-fertilization (dpf) *Tg(aldoca:TRPV1-tagRFP)* larvae. Scale bar 100 μ m. (C) Schematic of strategy for dose-dependent activation (yellow, left) or lesion (red, right) of Purkinje cells by addition of the TRP channel agonist capsaicin (Csn). (D) Confocal image (inverted look-up table) of one cerebellar hemisphere of *Tg(aldoca:TRPV1-tagRFP); Tg(elavl3:h2b-GCaMP6f)* larvae before, 3, 6, and 9 h after addition of capsaicin. Heart corresponds to the labelled trace in (E). (E) Normalized change in fluorescence following treatment with 1 μ M capsaicin in individual Purkinje cells as a function of time. Purkinje cells from *Tg(aldoca:TRPV1-tagRFP);Tg(elavl3:h2b-GCaMP6f)* larvae (orange) and *Tg(elavl3:h2b-GCaMP6f)* control larvae (grey). (F) Timelapse images of Purkinje cell axons in *Tg(aldoca:TRPV1-tagRFP)* larvae immediately after addition of 10 μ M capsaicin. Scale bar 10 μ m. (G) Confocal images of cerebellar hemispheres of *Tg(aldoca:TRPV1-tagRFP)* larvae before (7 dpf, left) and after (9 dpf, right) treatment with 10 μ M capsaicin. Control larvae (DMSO, top) and lesion larvae (10 μ M capsaicin, bottom). Scale bar 10 μ m. (H) Quantification of Purkinje cell numbers of fish (n=3) from (G).

Induced activation was reversible, even after prolonged exposure to 1 μM of capsaicin. We tested whether the elevated patterns of neuronal activity that we observed in the presence of capsaicin would return to baseline by imaging cerebellar Purkinje cells in *Tg(elavl3:h2B-GCaMP6f)* before exposure, after 6 h of 1 μM capsaicin, and 40 min after washout. Relative to baseline, fluorescent intensities increased after 6 h, as in **Figure 1E**. Importantly, fluorescence returned to baseline levels after 40 min of washout (**Figure S1B**, 6h post 1 μM capsaicin: 40.9%/TRPV1+ cells $F/F_0 > 2$; washout: 0%/ TRPV1+ cells $F/F_0 > 2$; 22 cells from 3 fish; activated cells before 1 μM capsaicin treatment vs. after washout: Fisher's exact test: $p = 1$). We conclude that capsaicin-induced activation is reversible after washout.

Exposure to high doses of capsaicin caused rapid axonal degeneration and cell death. We developed a protocol for Purkinje cell lesion: *Tg(aldoca:TRPV1-tagRFP)* larvae (without GCaMP6f) were imaged at 7 dpf, at 8 dpf after 1 h of 10 μM capsaicin treatment and again at 9 dpf (**Figure 1G**). Timelapse imaging of the Purkinje cell axons showed rapid degeneration already 15 min after capsaicin treatment started (**Figure 1F**). Cell numbers rapidly declined after 1 h of 10 μM capsaicin treatment and did not show any signs of recovery at 9 dpf (**Figures 1G** and **1H**) (median [inter-quartile range]; 7 dpf: control 213 [195 – 271] cells vs. pre lesion 282 [255 – 366] cells; 9 dpf: control 218 [201 – 250] cells vs. post lesion 68 [52 – 70] cells; 3 control and 3 lesioned fish).

Consistent with prior work in other cell populations⁶¹, we found that chemogenetic use of the capsaicin/TRPV1 system can be used to reversibly activate or rapidly ablate cerebellar Purkinje cells in larval zebrafish.

Purkinje cells regulate postural control in the pitch axis

We used our Scalable Apparatus to Measure Posture and Locomotion (SAMPL) to measure posture and locomotion in freely swimming zebrafish²⁶. SAMPL is a high-throughput videographic approach that measures kinematic parameters of posture and locomotion from fish swimming in a predominantly vertical arena that encourages navigation in depth (**Figures 2A** and **2B**) allowing us to analyze postural changes in the pitch (nose-up/nose-down) axis. Larval zebrafish locomote in discrete bouts of rapid translation (**Figure 2B**, grey lines). To navigate up/down, fish sequence these bouts while maintaining a nose-up/nose-down pitch⁶⁵. Notably, climb/dive bouts are defined relative to the *trajectory* of the bout. Climb/dive bouts can therefore be initiated from either nose-up (positive) or nose-down (negative) *postures*.

Nose-up “climb” bouts (**Figure 2E**) engage both axial musculature of the body and the fins to produce a net upward trajectory while nose-down “dive” bouts (**Figure 2J**) rely on axial musculature alone and have a net downward trajectory²⁵. Notably, postural angles after either climb or dive bouts tend to increase, a consequence of restorative rotations that counteract destabilizing torques²⁴. SAMPL's automated and high-throughput design yields data with large numbers of observations. To ensure a focus on only the most meaningful differences, we adopted two stringent criteria for significance: p -values < 0.05 , and an effect size of $\geq 15\%$. All p -values and effect sizes are reported in **Tables 1** to **5**.

We used the timing and capsaicin concentrations we had previously validated (**Figure 1**) to design two behavioral paradigms: one to activate and one to ablate cerebellar Purkinje cells. Experiments were done from 7–9 dpf, and began with a single day without perturbations. Activation was then achieved by exposing *Tg(aldoca:TRPV1-tagRFP)*; *Tg(elavl3:h2B-GCaMP6f)* fish to two 6h periods of 1 μM capsaicin while they swam freely in the dark (**Figure 2C**). Alternatively, Purkinje cells were ablated by exposing *Tg(aldoca:TRPV1-tagRFP)* fish to 10 μM of capsaicin for 1h (**Figure 2D**). All fish were screened before experiments for comparable levels of tagRFP fluorescence and control and experimental groups were randomly selected. A single experimental repeat consisted of 1-3 apparatus run in parallel with fish from a single clutch of embryos (i.e. siblings). To maintain consistency with genotypes used for validation, the activation

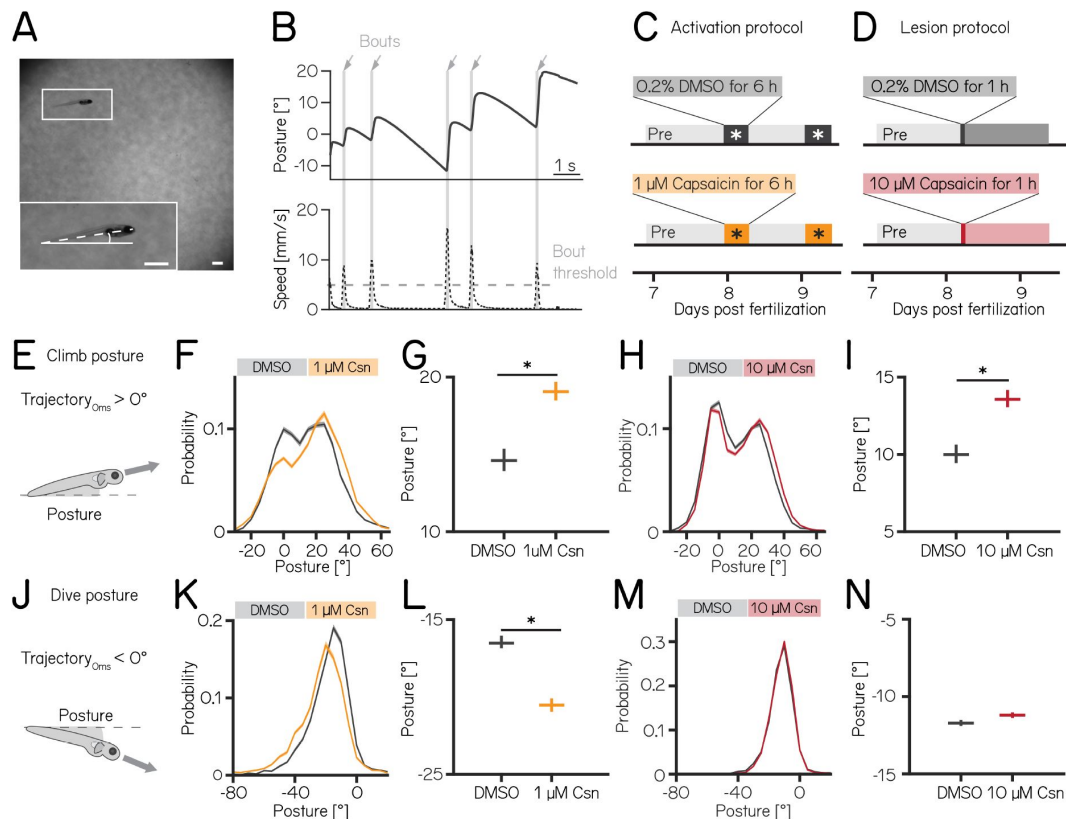


Figure 2

Both chemogenetic activation and ablation of Purkinje cells modify postural stability.

(A) Sample image of a freely-swimming zebrafish larva imaged from the side. Inset shows the larva at higher magnification view and its pitch, defined as the angle between the horizon (straight line) and the long axis of the body (dashed line). Scale bars 1mm. (B) Pitch angle (posture, top) and speed (bottom) as a function of time for one recorded epoch. Individual swim bouts (speed > 5 mm/s threshold) are highlighted in grey (arrows). (C) Timecourse for activation experiments between 7–9 dpf. Larvae received 1 μ M of capsaicin in 0.2% DMSO twice on days 8&9 for 6h each. (D) Timecourse for lesion experiments; larvae received a single dose of 10 μ M capsaicin in 0.2% DMSO for 1h on day 8. (E) Climbs are defined as a bout where the trajectory at peak speed took the fish nose-up ($>0^\circ$). (F) Probability distribution of climb postures for control (black) and 1 μ M capsaicin treated larvae (yellow). Data is shown as median and inter-quartile range. (G) Average climb posture of control and activated larvae (8 repeats/149 control fish; 8 repeats/155 1 μ M capsaicin treated fish; climb postures: 14.7° [$14.0 - 15.4^\circ$] vs. 19.0° [$18.5 - 19.7^\circ$], p-value < 0.001, effect size: 29%, Wilcoxon rank sum test). (H) Probability distribution of climb postures for control (black) and 10 μ M capsaicin treated larvae (red). Data is shown as median and inter-quartile range. (I) Average climb posture of control and lesioned larvae (14 repeats/110 control fish; 14 repeats/120 10 μ M capsaicin treated fish; climb postures: 10.0° [$9.5 - 10.7^\circ$] vs. 13.6° [$13.1 - 14.3^\circ$], p-value < 0.001, effect size: 36%, Wilcoxon rank sum test). (J–N) Same as E–I, but for dive bouts (trajectory that took the fish in the nose-down direction). (L) Average dive posture of control and activated larvae (8 repeats/149 control fish; 8 repeats/155 1 μ M capsaicin treated fish; dive postures: -16.6° [$-16.9 - -16.1^\circ$] vs. -20.5° [$-20.9 - -20.1^\circ$], p-value < 0.001, effect size = 24%, Wilcoxon rank sum test). (N) Average dive posture of control and lesioned larvae (14 repeats/110 control fish; 14 repeats/120 10 μ M capsaicin treated fish; dive postures: -11.7° [$-11.9 - -11.5^\circ$] vs. -11.2° [$-11.4 - -11.0^\circ$], p-value = 0.002, effect size = -4%, Wilcoxon rank sum test). Unless otherwise indicated data are shown as median with 95% confidence interval, * indicates p-value < 0.05 and effect size $\geq 15\%$

	control median [95% CI]	activation median [95% CI]	effect [%]	p-value	significance
pre activation					
critical p-value: 0.006					
climb posture [°]	15.4 [15.1 – 15.8]	16.1 [15.6 – 16.4]	4	0.006	no
dive posture [°]	-12.9 [-13.2 – -12.6]	-13.5 [-13.8 – -13.3]	5	<0.001	no
bout duration [s]	0.2 [0.2 – 0.2]	0.2 [0.2 – 0.2]	0	<0.001	no
Inter-bout interval [s]	1.4 [1.4 – 1.4]	1.4 [1.4 – 1.4]	2	<0.001	no
speed [mm/s]	12.5 [12.4 – 12.6]	12.6 [12.5 – 12.6]	0	0.831	no
slope slow [mm/s]	0.014 [0.014 – 0.014]	0.012 [0.012 – 0.012]	-4	0.078	no
slope medium [mm/s]	0.017 [0.016 – 0.017]	0.016 [0.016 – 0.016]	-1	0.352	no
slope fast [mm/s]	0.041 [0.041 – 0.041]	0.042 [0.041 – 0.042]	1	0.385	no
post activation					
critical p-value: 0.006					
climb posture [°]	14.7 [14.0 – 15.4]	19.0 [18.5 – 19.7]	29	<0.001	yes
dive posture [°]	-16.6 [-16.9 – -16.1]	-20.5 [-20.9 – -20.1]	24	<0.001	yes
bout duration [s]	0.2 [0.2 – 0.2]	0.2 [0.2 – 0.2]	0	<0.001	no
Inter-bout interval [s]	1.6 [1.5 – 1.6]	1.6 [1.5 – 1.6]	2	0.043	no
speed [mm/s]	12.7 [12.5 – 12.8]	13.0 [12.9 – 13.1]	3	<0.001	no
slope slow [mm/s]	0.012 [0.011 – 0.012]	0.009 [0.009 – 0.010]	-6	0.149	no
slope medium [mm/s]	0.019 [0.018 – 0.019]	0.015 [0.015 – 0.015]	-11	<0.001	no
slope fast [mm/s]	0.037 [0.037 – 0.038]	0.034 [0.034 – 0.035]	-8	0.069	no

Table 1.

Behavior measurements 7 dpf Purkinje cell activation.

and ablation experiments had different backgrounds (i.e. the presence/absence of the *elavl3:h2B-GCaMP6f* allele). Because of background variation²⁶, all comparisons were restricted to control vs. experimental groups *within* an experimental paradigm over the same time period. We focused our analysis on the pitch axis as the current version of the SAMPL apparatus²⁶ is not optimized for quantification of roll axis behavior. Across our datasets (**Tables 1** to **3**) we did not observe meaningful differences between the control and experimental groups in the pre-manipulation period. To avoid adding noise to our estimates of effect size, we therefore report comparisons between control and experimental groups after perturbation. We did not observe global consequences for swimming: swim speed, swim frequency and bout duration were unaffected during Purkinje cell activation (**Figures S2A** and **S2B**) or after Purkinje cell lesion. Similarly, raw bout numbers were not different between the control and activation (median [inter-quartile range] 1256 [571 – 1454] bouts vs. 656 [444 – 1485] bouts; p-value 0.46) or lesion groups (2004 [1507 – 2471] bouts vs. 1913 [1556 – 2416] bouts; p-value 0.84, **Tables 1** and **2**).

Climbing postures were perturbed after both activation and ablation of Purkinje cells. During activation, fish adopted more nose-up postures before and throughout climb bouts. We observed a shift towards more positive values across the distribution of postures before fish initiated a climb bout (**Figure 2F**). The average climb posture of fish during depolarization was 29% higher than in control fish (**Figure 2G**, median [95% confidence interval]: 14.7° [14.0 – 15.4°] vs. 19.0° [18.5 – 19.7°], p-value < 0.001, effect size: 29%). Similarly, after Purkinje cell lesion, the average climb bout postural angle increased 36% relative to controls (**Figures 2H** and **2I**, 10.0° [9.5 – 10.7°] vs. 13.6° [13.1 – 14.3°], p-value < 0.001, effect size: 36%).

We observed an unexpected decrease in the climb bout postural angles for control fish in the post-lesion period (from 18.0° [17.6 – 18.4°] to 10.0° [9.5 – 10.7°], **Table 2**). We do not have an explanation for this particular change and we have confirmed that it is not due to a single outlier experiment (detected outliers: 0/15 pre-lesion control experiments; 0/14 post-lesion control experiments). Notably, if we assess the effect of adding 10 µM capsaicin by comparing the magnitude of the relative difference between pre- and post-lesion periods, normalized to the pre-lesion period, we still see a significant difference (control vs. lesion: -46% vs. -26%). We conclude that, even when accounting for observed changes between control fish at 7 vs. 8 dpf, Purkinje cell ablation modifies climb postures.

Dive bout postures were similarly perturbed after activation, but not ablation of Purkinje cells. Fish adopted more nose-down postural angles before and throughout dive bouts with a leftward shift of the distribution of postures before dive bouts (**Figure 2K**). Average dive bout posture was 24% more negative than in control fish (**Figure 2L**, median [95% confidence interval]: -16.6° [-16.9 – -16.1°] vs. -20.5° [-20.9 – -20.1°], p-value < 0.001, effect size = 24%). Purkinje cell lesions at 7 dpf did not shift the average posture for dive bouts (**Figures 2M** and **2N**, -11.7° [-11.9 – -11.5°] vs. -11.2° [-11.4 – -11.0°], p-value = 0.002, effect size = -4%).

We interpret these data as evidence that Purkinje cell activity is crucial to ensure that posture during climbs and dives is maintained within a normal range.

Loss of Purkinje cells in older fish results in more global deficits to posture

Over the first two weeks of life, larval zebrafish morphology and postural control strategies develop considerably²³. These changes are matched by similarly pronounced cerebellar growth⁶⁶ (**Figures 3A** and **3B**). We observed that the number of Purkinje cells labelled in *Tg(aldoca:TRPV1-tagRFP)* roughly doubled between 7 and 14 dpf (**Figure 3C**, median [inter-quartile range] 7 dpf: 282 [255 – 336]; 14 dpf: 662 [591 – 733]). The increase in cell numbers is also evidence that the *aldoca* promoter continued to drive expression at later stages, allowing us to perform comparative experiments.

Table 2.

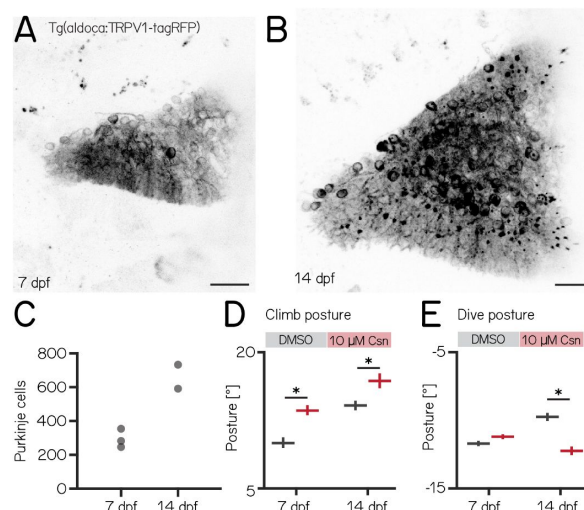
Behavior measurements 7 dpf Purkinje cell lesion.

	control median [95% CI]	lesion median [95% CI]	effect [%]	p-value	significance
pre lesion					
critical p-value: 0.006					
climb posture [°]	18.0 [17.6 – 18.4]	19.0 [18.6 – 19.4]	6	0.001	no
dive posture [°]	-11.9 [-12.1 – -11.6]	-11.5 [-11.7 – -11.3]	-3	0.25	no
bout duration [s]	0.2 [0.2 – 0.2]	0.2 [0.2 – 0.2]	4	0.001	no
Inter-bout interval [s]	1.8 [1.8 – 1.9]	1.7 [1.7 – 1.8]	-4	<0.001	no
speed [mm/s]	11.3 [11.2, 11.4]	12.0 [11.9, 12.1]	6	<0.001	no
slope slow [mm/°]	0.003 [0.003 – 0.003]	0.004 [0.004 – 0.004]	2	0.261	no
slope medium [mm/°]	0.008 [0.008 – 0.008]	0.001 [0.001 – 0.002]	-14	<0.001	no
slope fast [mm/°]	0.050 [0.049 – 0.050]	0.052 [0.051 – 0.053]	5	0.284	no
post lesion					
critical p-value: 0.006					
climb posture [°]	10.0 [9.5 – 10.7]	13.6 [13.1 – 14.3]	36	<0.001	yes
dive posture [°]	-11.7 [-11.9 – -11.5]	-11.2 [-11.4 – -11.0]	-4	0.002	no
bout duration [s]	0.2 [0.2 – 0.2]	0.1 [0.1 – 0.1]	-4	<0.001	no
Inter-bout interval [s]	1.7 [1.7 – 1.8]	1.7 [1.7 – 1.7]	-2	0.203	no
speed [mm/s]	10.3 [10.2 – 10.4]	10.6 [10.5 – 10.7]	2	<0.001	no
slope slow [mm/°]	0.008 [0.007 – 0.008]	0.005 [0.005 – 0.005]	-6	0.002	no
slope medium [mm/°]	0.012 [0.012 – 0.012]	0.008 [0.007 – 0.008]	-10	<0.001	no
slope fast [mm/°]	0.047 [0.047 – 0.048]	0.025 [0.025 – 0.026]	-46	<0.001	yes

Figure 3

Changes to postural stability after chemogenetic ablation of Purkinje cells are more pronounced in older fish.

(A) Confocal image of Purkinje cells in the cerebellum of a 7 dpf *Tg(aldoca:TRPV1-tagRFP)* larvae. Scale bar: 25 μ m. (B) Confocal image of Purkinje cells in the cerebellum of a 14 dpf *Tg(aldoca:TRPV1-tagRFP)* larvae. Scale bar: 25 μ m. (C) Increase in Purkinje cell numbers between 7 and 14 dpf. (D) Average climb bouts postures for 7 dpf control and lesion larvae (left) and 14 dpf control and lesion larvae (right). (14 dpf lesion: 7 repeats/48 control fish; 7 repeats/44 10 μ M capsaicin treated fish; climb postures: 14.3° [13.8 – 14.8°] vs. 17.1° [16.2 – 17.8°]; p-value < 0.001; effect size: 20%, Wilcoxon rank sum test). (E) Average dive bouts postures for 7 dpf control and lesion larvae (left) and 14 dpf control and lesion larvae (right). (14 dpf lesion: 7 repeats/48 control fish; 7 repeats/44 10 μ M capsaicin treated fish; dive postures: -9.8° [-10.1 – -9.5°] vs. -12.3° [-12.6 – -11.9°]; p-value < 0.001; effect size: 26%, Wilcoxon rank sum test). all data are shown as median with 95% confidence interval, * indicates p-value < 0.05 and effect size $\geq$ 15%



Similar to lesions at 7 dpf, we did not observe any differences in swim speed, frequency or bout duration (Table 3). At 14 dpf, the effects of Purkinje cell lesions on postural angles were more widespread than at 7 dpf and also affected dive postures. We repeated our previous ablation experiments (Figure 2D) between 14–16 dpf, and analyzed climb (Figures S3A and 3D) and dive bouts (Figures S3B and 3E). Loss of Purkinje cells created more widespread behavioral deficits. Specifically, climb bout posture was increased by 20% after Purkinje cell lesion (median [95% confidence interval]: 14.3° [13.8 – 14.8°] vs. 17.1° [16.2 – 17.8°]; p-value < 0.001; effect size: 20%). At 14 dpf we also observed an effect on dive bout postures. After lesion dive bouts postures were 26% more negative (-9.8° [-10.1 – -9.5°] vs. -12.3° [-12.6 – -11.9°]; p-value < 0.001; effect size: 26%).

We conclude that, consistent with morphological growth, Purkinje cells of the cerebellum play a broader role in postural control at 14 dpf than at younger ages.

Purkinje cells regulate speed-dependent fin engagement

To climb, larval zebrafish coordinate fin movements that generate lift with axial rotations that direct thrust (Figure 4A). The greater the axial rotation, the stronger the lift-producing fin movements; this relationship increases as larvae develop²⁵. Our previous work suggested that Purkinje cells were necessary for such fin-body coordination²⁵. Here, we observed that fin engagement is speed-dependent, with faster bouts producing greater lift for a given axial rotation (Figure 4B, left, Spearman's correlation coefficient: 0.2193; p = <0.001; of lift/rotation ratio [mm/deg] versus speed [mm/s] Figure S4A).

After Purkinje cell ablation, 14 dpf fish produced less lift than expected when they swam fast. We divided swim bouts into three different bins according to their peak speed (slow: 5–7.5 mm/s; medium: 7.5–15 mm/s; fast >15 mm/s) for both control and fish treated with 10 μM capsaicin. We parameterized the relationship between upward rotation and lift by fitting a line to swim bouts for each speed. After capsaicin exposure, the slopes of the medium and fast speed bins were significantly lower (Figures S4 and 4C), reflecting a loss of speed-dependent modulation (median [95% confidence interval]: slope slow: 0.029 [0.028 – 0.031 mm/°] vs. 0.033 [0.032 – 0.034 mm/°], p-value = 0.341, effect size: 6%; slope medium: 0.041 [0.040 – 0.044 mm/°] vs. 0.018 [0.018 – 0.019 mm/°], p-value < 0.001, effect size: -34%; slope fast: 0.068 [0.067 – 0.071 mm/°] vs. 0.026 [0.026 – 0.027 mm/°], p-value < 0.001, effect size: -62%). The correlation between speed and fin-lift / rotation ratio is reduced after capsaicin exposure (Figure S4; Spearman correlation coefficient: control 0.2193; 10 μM capsaicin: 0.0397; Z-test after z-transformation: p < 0.001). When analyzing fin lift and upward rotation across the three speed bins separately, we only observed significant differences for fast swim bouts, specifically showing a reduction in fin lift and an increase in upward rotation Table 4.

Next, to determine if lift was fin-dependent, we amputated the fins^{25,26} and repeated our experiments. A detailed explanation of how fin amputation affects swim kinematics can be found here^{25,26}. We observed a near total loss of lift at all speeds; regardless of the speed bin, the slope of the relationship between upward rotation and lift was indistinguishable from zero (slope slow: 0.036 [0.035 – 0.037 mm/°] vs. -0.005 [-0.005 – -0.004 mm/°], p-value < 0.001; effect size: -60%; slope medium: 0.055 [0.054 – 0.056 mm/°] vs. -0.005 [-0.005 – -0.005 mm/°], p-value < 0.001; effect size: -88%; slope fast: 0.068 [0.067 – 0.069 mm/°] vs. 0.013 [0.013 – 0.013 mm/°], p-value < 0.001; effect size: -81%). Finally, we examined fin-body coordination in our 7 dpf activation and ablation datasets. In contrast to older larvae, we observed no meaningful changes after activation of Purkinje cells at 7 dpf. For Purkinje cell lesions at 7 dpf we found only the fin body coordination at fast bouts to be affected Tables 1 and 2.

Our data show that loss of Purkinje cells disrupts the speed-dependent increase in fin-mediated lift in older, and to a lesser degree in younger fish. We interpret this finding as evidence that Purkinje cells are indispensable for normal coordination of the fins and body.

Table 3.

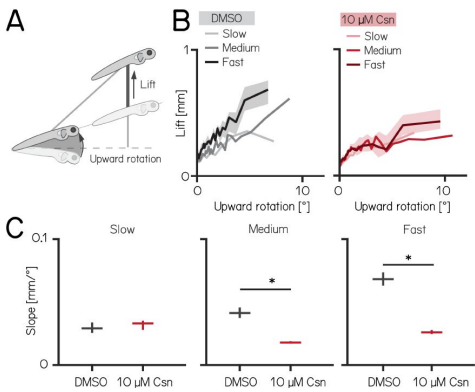
Behavior measurements 14 dpf Purkinje cell lesion.

	control median [95% CI]	lesion median [95% CI]	effect [%]	p-value	significance
pre lesion					
critical p-value: 0.01					
climb posture [°]	16.0 [15.6 – 16.5]	15.2 [14.7 – 15.9]	-5	<0.001	no
dive posture [°]	-9.0 [-9.4 – -8.7]	-9.5 [-9.7 – -9.1]	5	0.006	no
bout duration [s]	0.2 [0.2 – 0.2]	0.2 [0.2 – 0.2]	3	0.59	no
Inter-bout interval [s]	2.3 [2.3 – 2.4]	2.4 [2.4 – 2.4]	4	0.29	no
speed [mm/s]	10.2 [10.0 – 10.3]	10.5 [10.4 – 10.7]	4	<0.001	no
post lesion					
critical p-value: 0.01					
climb posture [°]	14.3 [13.8 – 14.8]	17.1 [16.2 – 17.8]	20	<0.001	yes
dive posture [°]	-9.8 [-10.1 – -9.5]	-12.3 [-12.6 – -11.9]	26	<0.001	yes
bout duration [s]	0.2 [0.2 – 0.2]	0.2 [0.2 – 0.2]	-8	<0.001	no
Inter-bout interval [s]	2.9 [2.8 – 3.0]	2.8 [2.7 – 2.8]	-4	0.01	no
speed [mm/s]	9.7 [9.6 – 9.8]	9.3 [9.2 – 9.4]	-3	<0.001	no

Figure 4

Chemogenetic ablation of Purkinje cells disrupts fin-body coordination in a speed-dependent manner.

(A) Larval zebrafish use two independent effectors (trunk and body) to climb. The contribution of each effector can be dissociated by the observed kinematics: changes to the angle of the trunk predict a trajectory for a particular bout (upward rotation). The actual position of the fish in depth at the end of the bout reveals the lift generated by the fins. A detailed kinematic examination of climbing, including fin ablations, is detailed in [25](#). (B) Coordination of fin and trunk engagement plotted as upward rotation against lift. Positive slopes reveal that larger rotations are coupled to greater fin engagement and greater changes in depth. The slope of this relationship becomes steeper for bouts with greater translational speed. Bouts from control (grey, left) and 10 μ M capsaicin treated larvae (red, right) are plotted at different swim speeds, shaded areas indicate 95% confidence interval of the median of the fast swim speeds. (C) Average slopes of lift/rotation curves for control and 10 μ M capsaicin treated larvae at different swim speeds. (8 repeats/15 control fish; 8 repeats/18 10 μ M capsaicin treated fish); slow: $p = 0.341$; medium: $p < 0.001$; fast: $p < 0.001$. Data are plotted as median with inter-quartile range. * indicates $p < 0.05$ and effect size $\geq 15\%$



	control	median [95% CI]	lesion	median [95% CI]	effect [%]	p-value	significance
lesion							
critical p-value: 0.004							
slope slow [mm/s]	0.029	[0.028 – 0.031]	0.033	[0.032 – 0.034]	6	0.341	no
slope medium [mm/s]	0.041	[0.040 – 0.044]	0.018	[0.018 – 0.019]	-34	<0.001	yes
slope fast [mm/s]	0.068	[0.067 – 0.071]	0.026	[0.026 – 0.027]	-62	<0.001	yes
R ² slow	0.212	[0.201 – 0.227]	0.371	[0.361 – 0.392]	35	0.005	no
R ² medium	0.382	[0.373 – 0.388]	0.603	[0.594 – 0.611]	48	<0.001	yes
R ² fast	0.460	[0.452 – 0.471]	0.641	[0.634 – 0.648]	39	0.001	yes
Fin lift slow [mm]	0.184	[0.168 – 0.199]	0.164	[0.155 – 0.172]	-11	0.004	no
Fin lift medium [mm]	0.201	[0.192 – 0.208]	0.178	[0.171 – 0.185]	-11	0.002	no
Fin lift fast [mm]	0.267	[0.258 – 0.277]	0.189	[0.174 – 0.197]	-29	<0.001	yes
Rotation slow [°]	1.471	[1.348 – 1.686]	1.254	[1.140 – 1.383]	-15	0.005	no
Rotation medium [°]	1.265	[1.176 – 1.324]	1.178	[1.120 – 1.275]	-7	0.515	no
Rotation fast [°]	1.083	[1.018 – 1.150]	1.241	[1.180 – 1.336]	15	<0.001	no

Table 4.

Behavior measurements 14 dpf Purkinje cell lesion.

Purkinje cells encode pitch direction at both individual and population levels

Our experiments establish that manipulations of Purkinje cells interfere with balance in the pitch axis. We therefore hypothesized that Purkinje cell activity would be modulated by nose-up/nose-down body tilts. We used Tilt In Place Microscopy (TIPM) ²⁶ to measure the response of individual Purkinje cells (**Figure 5A**) to rapid pitch tilts. Briefly, fish are mounted on a mirror galvanometer and rapidly rotated to eccentric angles (**Figure 5B**, $\pm 30^\circ$ nose-up/nose-down).

We used *Tg(aldoca:GAL4);Tg(UAS:GCaMP6s)* to label Purkinje cells in the lateral parts of the cerebellum thought to receive vestibular input ^{30,39,67}. To facilitate identification of the same cells from volumes imaged at both $\pm 30^\circ$, we used doubly mono-allelic fish and screened for sparse expression of Purkinje cells. In total, we imaged 43 Purkinje cells from 8 fish. Of those, 31 cells could reliably identified at $\pm 30^\circ$ and were included in the analysis.

We calculated a directionality index (DI) for all cells and categorized cells as either tuned ($|DI| > 0.35$) or as untuned ($|DI| < 0.35$). Some cells showed, on average, higher responses to one direction (**Figure 5F**), but due to highly variable responses (median [inter-quartile range] Up response range: 23.3 [18.3 – 29.5]; Down response range: 43 [23.7 – 49.2]), they did not exhibit consistent tuning as determined by the directionality index. Individual Purkinje cells showed either directionally-tuned (**Figure 5C**, $n=18$) or untuned (**Figure 5D**, $n=13$) patterns of responses. Tuned cells were distributed throughout the lateral cerebellum (**Figure 5E**), and showed a slight preference for nose-down stimuli (12 vs. 6, **Figure 5F**). We did not observe any systematic differences in the response properties across each experiment from untuned cells (**Figure 5G**).

While untuned cells did not show overt directional preferences, pooling their responses allowed decoding of stimulus direction. We applied principal component analysis (PCA) to visualize the data and assess whether the trial types even for untuned cells exhibited distinct separation based on their trial identity (nose-up or nose-down). Principal component analysis of the integral of the full responses on each trial from untuned neurons showed near-complete segregation of trial types (**Figure 5H**). To assay whether there was indeed directional information we trained a decoder (support vector machine) and tested its accuracy on pseudo-populations of untuned cells of different sizes ranging from 3 - 13 cells (**Figure 5I**). We focused only on untuned cells, as including even a single tuned cell for the population coding will lead to excellent results. Training and test trials were different to avoid over-fitting. Pseudo-populations with more than 3 cells achieved accurate decoding well above chance levels (determined by shuffling trial identity) (median [inter-quartile range] accuracy: 3/5/7/10/13 cells: 0.78 [0.68 – 0.91] / 0.88 [0.70 – 0.88] / 1 [0.84 – 1] / 1 [0.97 – 1] / 1 [1 – 1]). To ensure that the choice of cutoff for the directionality index (0.35) does not bias the decoding results, we also calculated the decoding accuracy using the 3, 5, and 7 least directionally tuned cells based on their directionality index. We again found that more than 3 cells achieve accurate decoding above chance level (3/5/7 least tuned cells: 0.67 [0.64 – 0.78] / 0.86 [0.73 – 0.88] / 0.88 [0.86 – 0.92]).

Older larvae showed additional changes to dive postures after Purkinje cell lesions. We therefore tested if: (1) Purkinje cells in older larvae exhibited differences in the numbers or direction of tuned cells or (2) if population-level decoding accuracy changed. We performed longitudinal TIPM, sampling from zebrafish larvae at 7 and 14 dpf. To improve throughput, we recorded the responses upon return from $\pm 19^\circ$ stimuli (**Figure S5A**) (7 dpf: 138/11 cells/fish; 14 dpf: 90/7 cells/fish; of those 23/3 cells/fish were imaged at both timepoints); previous work established that responses upon return to baseline are highly correlated with the response at the eccentric position ⁶⁸. We observed increased fluorescence relative to baseline values in 7 dpf and 14 dpf Purkinje cells upon return from $\pm 19^\circ$ steps (**Figure S5B** and **Figure S5C**). To analyze directional tuning we compared the maximum fluorescence in the first second after return to baseline. The relative

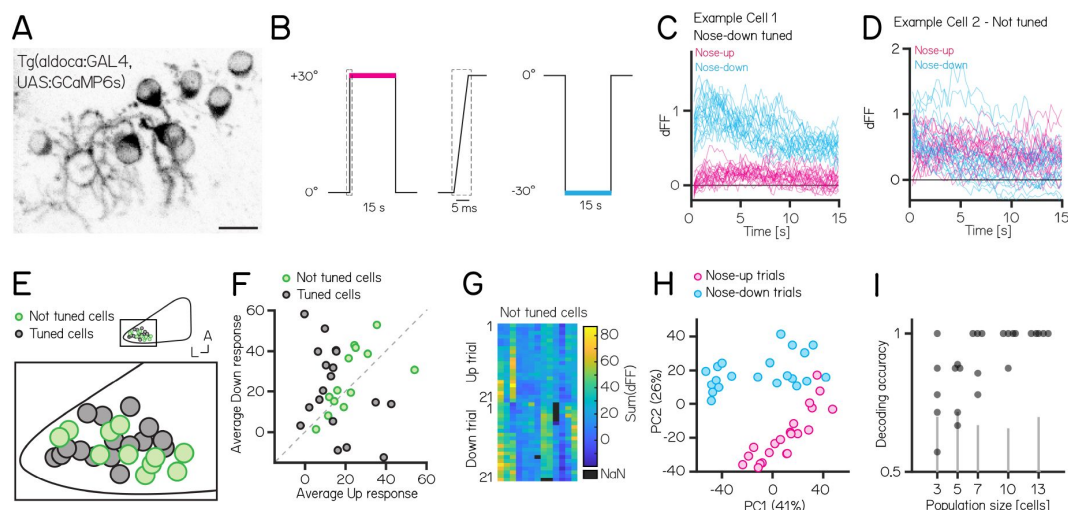


Figure 5

Activity in larval zebrafish Purkinje cells can differentiate nose-up from nose-down pitch both individually and collectively.

(A) 2-photon image of Purkinje cell somata expressing a calcium indicator in the *Tg(aldoca:GAL4);Tg(UAS:GCaMP6s)* line. Scale bar 10 μ m. (B) Pitch tilt stimuli consisted of rapid galvanometer steps for 15 seconds in the nose up (+30°, pink) and nose-down (-30°, blue) direction. Inset in dotted rectangle shows the near-instantaneous timecourse of the step. (C) Example responses (n=42) from a single Purkinje cell sensitive to nose-down pitch (blue) but not nose-up (pink). (D) Example responses (n=42) from a single Purkinje cell without directional selectivity. (E) Superimposed positions of Purkinje cell somata within a single cerebellar hemisphere; no obvious topography separates tuned (black, n=16) and untuned (green, n=11 |directionality index| < 0.35) cells. (F) Averaged integrated response (dFF) for individual cells over the 15 second stimulus plotted for nose-up vs. nose-down stimuli, colored by tuned (black) and untuned (green). (G) Heatmap of integrated response (dFF) for 13 untuned neurons on 21 up/down tilts. (H) Principal component analysis of integrated responses for untuned neurons for each of 21 up (pink) and 21 down (blue) trials. (Percentage of variance explained) (I) Performance of a support vector machine for binary classification of up/down tilt using integrated responses from increasing numbers of untuned neurons. Dots are different sets of neurons, gray lines shows the spread of performance from shuffled up/down identity (median [interquartile range] accuracy: 3/5/7/10/13 cells: 0.78 [0.68 – 0.91] / 0.88 [0.70 – 0.88] / 1 [0.84 – 1] / 1 [0.97 – 1] / 1 [1 – 1]).

number of tuned cells per fish was comparable between 7 and 14 dpf larvae (**Figure S5D** [↗](#); median [inter-quartile range] at 7 dpf: 7 [6 – 34%]; 14 dpf: 8 [2 – 19%]; p-value = 0.7763). While most cells were not directionally selective, the preferred direction of tuned cells was different at 7 and 14 dpf: at 7 dpf more Purkinje cells were nose-down tuned (2/31 up/down) but at 14 dpf more cells were nose-up tuned (11/3 up/down **Figure S5E** [↗](#); Fisher's exact test, p-value < 0.001).

We next assayed accuracy of directional encoding of simultaneously recorded untuned cells. We performed principal component analysis for untuned cells at 7 (**Figure S5F** [↗](#)) and 14 dpf (**Figure S5G** [↗](#)) and tested decoding accuracy on the untuned cells of individual fish at 7 and 14 dpf. We did not observe differences in decoding accuracy between 7 and 14 dpf larvae (**Figure S5H** [↗](#); median [inter-quartile range] 7 dpf: 0.68 [0.63 – 0.83]; 14 dpf: 0.73 [0.65 – 0.79]; p-value = 0.9468). We conclude that cerebellar Purkinje cells can encode pitch direction both at the single neuron and population levels with similar encoding accuracy in young and older larvae.

Discussion

We used a validated chemogenetic tool to investigate the role of cerebellar Purkinje cells in postural behavior as larval zebrafish swam freely in depth. Activation of Purkinje cells could induce changes in pitch axis (nose-up/nose-down) posture. Purkinje cell ablation changed posture, with broader effects in older larvae. Ablation disrupted fin-body coordination responsible for proper climbing. Finally, we could reliably decode pitch-tilt direction from patterns of Purkinje cell activity. We did not observe developmental changes in population coding of direction but found a shift in the tuning direction of Purkinje cells. Taken together our results establish a role for the cerebellum in postural control even during the earliest stages of larval zebrafish development. Our work establishes a new method that combines bidirectional manipulation of cerebellar output and quantitative high-throughput measures of unconstrained posture and locomotion.

Contributions of Purkinje cells to posture

While activation and ablation manipulations both produced biologically meaningful behavior changes, the two experiments were run with different genetic backgrounds and on different generations of the SAMPL apparatus. Consequently, our ability to define precisely what role Purkinje cells play in balance behaviors in larval zebrafish is limited. Activation experiments are particularly laborious as they require thorough pre-screening to ensure adequate brightness levels to achieve sufficient depolarization without excitotoxicity. To maintain consistency in depolarization effects, experiments involving different genetic backgrounds or developmental stages would require additional calcium imaging to confirm that 1 μ M capsaicin elicits comparable responses. Hence, we restricted our experiments to 7 days post-fertilization (dpf) and did not extend the study to other developmental time points. Given that the primary purpose of this series of experiments was to establish TRPV1-mediated manipulation of Purkinje cells as a means to investigate postural control, it is beyond the scope of the work to repeat the experiments. Nonetheless, we consider the findings individually below in the context of prior work.

Purkinje cell ablations modifies postural stability. Importantly, the differences we observed were more widespread in older larvae, underscoring the developmental importance of Purkinje cells for balance. Purkinje cell output is inhibitory ⁶⁹[↗](#),⁷⁰[↗](#), Purkinje cells in the lateral cerebellum project to vestibular nuclei ³⁰[↗](#),³⁹[↗](#), and Purkinje cells are tonically active ⁷¹[↗](#),⁷²[↗](#). We propose that the net effect of Purkinje cell loss would be disinhibition of target nuclei responsible for encoding posture and parameterizing corrective pitch-axis behaviors. While the precise nature of the transformation between larval zebrafish pitch and posture control kinematics is not yet known, loss of cerebellar-targeted nuclei can disrupt postural behaviors ²⁵[↗](#),⁷³[↗](#).

The effects of ablations became more widespread in older larvae. During early development, larval zebrafish grow in volume by roughly an order of magnitude and shift their postural control strategies to better climb/dive as they navigate in depth ^{23,25}. Unlike climb bouts, changes to postural stability during dives only emerge at 14 dpf. As activation of Purkinje cells produced meaningful changes during dives at 7 dpf, we infer that the delayed emergence of ablation effects does not reflect incomplete integration of Purkinje cells into dive-control circuits. Instead, we propose that this delay reflects the gradual, functional emergence of Purkinje cell control over dives as development progresses. This aligns with previous studies showing that Purkinje cell circuits mature at different rates during development ¹⁴.

Notably, the basal posture during dive bouts decreases in older control fish (**Figure 3E**) — ablation shifts the posture comparable to its younger state. Future work with our system enables testing of the hypothesis that Purkinje cell output plays a role in setting the postures older fish adopt during dives.

Purkinje cell activation also modifies postural stability. Intriguingly, activation broadened the distribution of observed postures in the same way as ablation. Our imaging assay established that 1 μ M of capsaicin would stochastically activate subsets of Purkinje cells. This stochasticity could reflect normal fluctuations in basal levels of activity, or it could arise from cells going in and out of depolarization block ⁷⁴. Synchronized/precisely-timed Purkinje cell output is thought to shape movements ^{75–80}, though perhaps not for all behaviors ⁸¹. Our imaging suggests that the set of Purkinje cells activated at any one moment in time is limited and random. We therefore propose that the net effect of 1 μ M of capsaicin is ultimately disruptive to Purkinje cell synchrony, and thus likely disruptive. Future work could test this hypothesis by intracellular recording from cerebello-recipient populations like the vestibular nuclei ^{82,83}.

Previously, we reported that larval zebrafish coordinate their fins and trunk to climb effectively ²⁵. The relationship between trunk-mediated changes to trajectory (upward rotation) and fin-mediated lift depends on locomotor speed. Here we observed that after Purkinje cell loss, speed-dependent increases in lift with greater trunk rotation are disrupted (**Figures S4A** to **4C**). In the fastest speed bin, we observed an increase in upward rotation and a decrease in average fin lift. In contrast, the medium-speed bin showed no significant changes in average fin lift or upward rotation, yet already displayed coordination deficits. Based on these observations, we argue that Purkinje cell lesions primarily affect coordination, rather than simply reducing one specific parameter such as lift or rotation. As we did not observe any change to locomotor speed after ablation (**Tables 1** to **3**), we infer that Purkinje cell loss disrupts speed-dependent coordination for climbing. These results extend our original report where a lower-throughput method (photoablation) suggested that Purkinje cell loss impacted the fin-trunk relationship ²⁵. In larval zebrafish, the neuronal substrates for axial speed control ^{84–89} and fin engagement ⁹⁰ are known. The potential for whole-brain imaging in larval zebrafish ⁴¹, particularly with high-speed voltage indicators ⁹¹ and cutting-edge modeling approaches ⁹², stands to reveal how Purkinje cell activity comes to coordinate body and fin movements. Importantly, since our behavioral data suggest that Purkinje cell activity impacts fin-trunk coordination more strongly in older larvae, longitudinal approaches will be key to understanding the developmental changes to cerebellar signaling that underlie effective coordination of trunk and limbs.

Encoding strategies for body tilt stimuli

Purkinje cell activity reflects both sensory and motor inputs. One limitation of TIPM is that larvae are immobilized in agarose during tilts. Consequentially, our measurements of Purkinje cell activity are artificially constrained. Nonetheless, a subset of Purkinje cells were unambiguously direction-selective, and a simple decoder could use the activity of non-selective cells to differentiate tilt direction. We infer that vestibular information directly related to pitch axis posture is represented by the Purkinje cell population targeted in our ablation/activation experiments, consistent with broader imaging of cerebellar responses to body tilt ^{45,46}.

Similar to the behavior results, we observed an asymmetry in the tuning direction of Purkinje cells at 7 dpf, with more cells being tuned to the nose-down direction. This asymmetry shifted between 7 and 14 dpf, suggesting developmental changes in how navigation in the pitch axis is processed in the cerebellum. These changes underscore the importance of longitudinal measurements of Purkinje cell activity across early development to understand emergent control of posture.

The ability to decode tilt direction from the collective activity of “untuned” Purkinje cells suggests a role for population coding. Such mechanisms have been proposed for head/body motion⁹³ and eye movements^{81,94} in the primate cerebellum. Population coding requires that multiple Purkinje cells converge onto downstream targets, which is well-established in cerebellar target nuclei^{76,95}. In larval zebrafish, Purkinje cells involved in locomotion converge on eurydendroid cells; electrophysiological recordings confirm a many-to-one convergence scheme that could similarly support population coding³¹. Vestibular-sensitive cells are located in the lateral cerebellum^{45,46}, which projects to hindbrain regions that contain vestibular nuclei⁶⁶. Comparing activity of vestibular nucleus neurons involved in tilt-driven behaviors^{82,96,97} before/after TRPV1-mediated ablation would speak to the collective contributions of Purkinje cells.

TRPV1/capsaicin as a tool to study cerebellar contributions to behavior

Our use of TRPV1/capsaicin complements a modern suite of molecular tools to target cerebellar Purkinje cells⁴⁸. In fish, different experiments have used opsins to excite / inhibit cerebellar Purkinje cells with exceptional temporal precision, establishing functional topography³⁰ and an instructive role in learning³¹. TRPV1/capsaicin is a well-validated approach⁶¹ that permits parametric (i.e. dose-dependent) activation/ablation with a single transgenic line. It does not require light, facilitating dissociation of vestibular from visual contributions without requiring genetically-blind fish as in other studies using excitatory opsins⁹⁸. Additionally, our approach of using TRPV1/capsaicin for cell ablation offers multiple advantages compared to other lesion methods. For example, Killer Red requires extended exposure to high light intensities and mounting of single fish²⁵. Nitroreductase ablation, while effective, requires extended pro-drug exposure, leading to slower and less precise results⁹⁹.

In contrast, TRPV1-mediated ablation is rapid and precise, with capsaicin triggering almost instantaneous cell death. This speed and simplicity make TRPV1 superior for experiments requiring quick, controlled ablation without the delays associated with other methods. Finally, chemogenetic approaches such as TRPV1/capsaicin permit prolonged experimentation in freely-moving animals, allowing us to collect large kinematic datasets necessary to rigorously study posture and locomotion.

Considerable progress has been made in recent years using new tools^{8,15,18} and new perspectives¹⁰⁰ to understand the cerebellar contributions to sensorimotor^{49–52} and non-sensorimotor behaviors^{53–56} in health and disease^{57–59}. Underlying this considerable progress is an ever-improving ability to manipulate the cerebellum without compromising rigorous measures of behavior. Here — in support of similar goals — we validated a new chemogenetic approach (TRPV1/capsaicin-mediated activation and ablation) compatible with a high-throughput paradigm to measure behavior in freely swimming larval zebrafish (SAMPL). Our data uncover expected signatures of cerebellar contributions to posture and coordination, establishing the validity of our approach. Further, by comparing the impact of Purkinje cell ablation in time, we leverage the rapid maturation of the zebrafish to open a window into cerebellar control of posture and coordination across development. Our approach establishes a path forward for the larval zebrafish model to contribute to cerebellar mechanisms of postural

control. The cerebellum emerged early in the evolution of vertebrates, when vertebrate life was underwater. Our work establishes a new tool to investigate ancient organizing principles of cerebellar function.

Materials and methods

Fish Care

All procedures involving zebrafish larvae (*Danio rerio*) were approved by the Institutional Animal Care and Use Committee of New York University. Fertilized eggs were collected and maintained at 28.5°C on a standard 14/10 hour light/dark cycle. Before 5 dpf, larvae were maintained at densities of 20-50 larvae per petri dish of 10 cm diameter, filled with 25-40 mL E3 with 0.5 ppm methylene blue. After 5 dpf, larvae were maintained at densities under 20 larvae per petri dish and fed cultured rotifers (Reed Mariculture) daily.

Fish Lines

To generate the *Tg(aldoca:TRPV1-TagRFP)* line, the 5-kbp aldolase Ca (aldoca) promoter⁶⁴ and a gene cassette that includes TRPV1-Tag1RFP cDNA, rabbit beta-globin intron, and the SV40 polyadenylation signal (pAS) in pT2-4xUAS:TRPV1-RFP⁶¹ were subcloned into the Tol2 vector pT2KDest-RfaF¹⁰¹ by the Gateway system (pT2K-aldoca-TRPV1-Tag1RFP-pAS). To establish stable transgenic lines, Tol2 plasmid and transposase mRNA (25 ng/μl each) were injected into one-cell-stage embryos.

The resulting *Tg(aldoca:TRPV1-tagRFP)* stable line allowed us to express the mammalian capsaicin-sensitive cation channel TRPV1 and the red fluorophore tagRFP in cerebellar Purkinje cells. Before exposure to capsaicin, fish were screened to ensure similar levels of tagRFP expression. When screening for transgene expression, we selected fish with clearly visible expression that was not excessively bright. The same criteria were applied when screening fish for GCaMP imaging and behavior experiments. Approximately a quarter of the fish that had aldoca:TRPV1-tagRFP expression had suitable expression levels for the activation experiment.

We measured neuronal activity using a genetically-encoded calcium indicator, *Tg(UAS:GCaMP6s)*⁸⁶, driven by *Tg(aldoca:GAL4)*¹⁰², or the *Tg(elavl3:h2B-GCaMP6f)* line¹⁰³.

Confocal imaging of TRPV1-mediated activation / lesion

Images were collected using a Zeiss LSM800 confocal microscope using a 20x 1.0NA water immersion objective. Larvae were mounted in 2% low melting point agar (catalog #16520, Thermo Fisher Scientific) in a dorsal up position. Anatomical images were acquired from fish anesthetized with 0.2 mg/ml ethyl-3-aminobenzoic acid ethyl ester (MESAB, catalog # E10521, Sigma-Aldrich). To activate TRPV1-expressing Purkinje cells, fish were treated with 1 μM capsaicin in 0.2% DMSO in E3. To lesion Purkinje cells, fish were exposed to 10 μM capsaicin in 0.2% DMSO in E3. Control fish were treated with 0.2% DMSO in E3. Agar was removed around the tip of the tail to facilitate drug delivery. Fish were mounted throughout functional imaging experiments and kept in temperature controlled incubators between timepoints. Time series images were acquired with a 2 photon microscope (Thorlabs Bergamo equipped with a Mai Tai HP laser tuned to 920nm) with a framerate of 7.9fps. Images were analyzed in Fiji¹⁰⁴, ROIs were drawn on nuclei of randomly selected Purkinje cells, which were then re-identified at each time point. Fluorescence for each cell and time point was normalized to the pre-capsaicin value. Cells with a dF/F0 of > 2 were considered activated and analyzed at the pre capsaicin and +6h timepoint. A Fisher's exact test was performed to determine significance of the activated cells.

To image the anatomy of Purkinje cells exposed to 10 μ M of capsaicin across time, the cerebellum was imaged at 7 dpf from fish mounted as above. Fish were unmounted and kept in E3 medium until the next day (8 dpf). At 8 dpf, fish were placed in 0.2% DMSO in E3 (control) or 10 μ M capsaicin in 0.2% DMSO in E3 for 40-60min, and imaged again after 1h of recovery in E3 post-treatment. Fish from both groups were imaged again at 9 dpf. Confocal images were analyzed in Fiji and Purkinje cell somata were counted in both hemispheres of the cerebellum. A conservative approach was taken for cell counting, with inclusion of any structures still resembling cells, regardless of potential non-functionality or signs of degradation. Consequently, the counts are likely an underestimate of the actual percentage of cell loss.

Zebrafish behavior recordings

All behavior was measured using the Scalable Apparatus for Measuring Posture and Locomotion (SAMPL) apparatus, consisting of a chamber where larvae could swim freely, an infrared illuminator, a camera, and software to process video in real time. A comprehensive description of the apparatus is contained in [26](#). Here we briefly describe the specific details of our experiments. Larvae were transferred to chambers at densities of 3-8 fish per chamber for 7 dpf experiments or 1-4 fish per chamber for 14 dpf experiments containing 25-30 ml of E3 or 0.2% DMSO / 1 μ M capsaicin for activation experiments. After 24 h, behavior recordings were paused for 30-60 minutes for feeding (feeding pause) and 1-2 ml of rotifer culture was added to each chamber. Larvae were removed from the apparatus after 48h.

To monitor behavior before/during Purkinje cell activation, 7 dpf larvae were placed in chambers with E3. At 8 & 9 dpf, control fish were placed in 0.2% DMSO in E3 and the condition fish were placed in 1 μ M capsaicin in 0.2% DMSO in E3 for 6h. The recording started about 10-15min after adding the fish to the capsaicin solution. Fish were fed after the 6h activation period. Video was sampled at 40Hz in constant darkness. Control: 9626 bouts (63% climb bouts)/149 fish/8 experimental repeats; Activation: 9664 bouts (61% climb bouts)/155 fish/8 experimental repeats;

To monitor behavior before/after Purkinje cell lesions, 7 dpf/14 dpf larvae were placed in the chambers with E3. After feeding at 8 dpf/15 dpf, fish were placed in petri dishes with 0.2% DMSO in E3 (control) or 10 μ M capsaicin in 0.2% DMSO in E3 for 40-60min. Fish were then returned to the chambers in E3 and behavior recording was started. Video was sampled at 160Hz in constant darkness. 7 dpf lesions: Control: 17895 bouts (61% climb bouts)/110 fish/14 experimental repeats; Lesion: 17819 bouts (57% climb bouts)/120 fish/14 experimental repeats; 14 dpf lesion: Control: 10666 bouts (58% climb bouts)/48 fish/7 experimental repeats; Lesion: 10708 bouts (54% climb bouts)/44 fish/7 experimental repeats;

Pectoral fin amputations were performed at 13 dpf. Two length-matched siblings were anesthetized in 0.2 mg/ml ethyl-3-aminobenzoic acid ethyl ester (MESAB, catalog # E10521, Sigma-Aldrich) simultaneously and mounted in 2% low-melting temperature agar. Visualized under a stereomicroscope (Leica M80, 20x/12 eyepieces, 1.0x objective), the two pectoral fins from one larva were removed by pulling the base of the fin at the scapulocoracoid laterally with #5 Dumont forceps. After amputation, both fish were freed from the agar and allowed to recover in E3 until the next day, at which point half of the amputated and control fish were randomly selected for Purkinje cell lesions. Lesions were performed as above and behavior recorded for 48h. Behavior was recorded at a sampling rate of 160Hz with a 14/10h light-dark cycle. Control: 1506/5090/5353 (slow/medium/fast) bouts/15 fish/8 experimental repeats; Purkinje cell lesion: 1667/6166/4299 (slow/medium/fast) bouts/18 fish/8 experimental repeats; Fin amputation: 1935/6295/4911 (slow/medium/fast) bouts/17 fish/8 experimental repeats.

Behavior analysis

Comprehensive descriptions of behavioral kinematics and baseline data for different genetic backgrounds are detailed in [26](#). Here we describe the specific parameters used for our experiments. Behavior data were analyzed using custom-written software in MATLAB (Mathworks, Natick MA), which extracted individual swim bouts from the raw data (x/z position and pitch angle as a function of time). Only bouts during the circadian day were analyzed. Experimental repeats consisted of data collected across multiple SAMPL boxes from a single clutch of fish; the number of fish available determined how many apparatus were used (1-3). For comparisons across conditions (e.g. activation/control), fish from one clutch were randomly split into control and condition groups. As bout number is the fundamental unit of kinematic analysis, and different numbers of fish available would yield different numbers of bouts, we bounded our experiments to allow comparison across repeats. Specifically, if an experimental repeat contained less than 650 bouts it was excluded.

Between 22-27% of lesion experimental repeats contained less than 650 bouts and were not included in the analysis. For the activation experiments 56% (10 of 18) of experimental repeats were excluded with the 650 bouts threshold due to shorter recording times a higher fraction of experiments contained less than the threshold number of bouts. In subsequent analyses, the number of analyzed bouts was matched from both groups for a given experimental repeat to ensure an identical representation of control and condition bouts. Individual bouts were aligned at the time of peak speed. Bouts were excluded if their peak speed was <5mm/s or the fish rotated more than 30°(120°/sec) during the acceleration. The fractions excluded were as follows: for 7 dpf ablation: ctrl 0.2% lesion 0.15%; 7 dpf activation: ctrl 1% activation 1.7%; 14 dpf ablation dark: ctrl 0.05% ablation 0.05%; 14 dpf ablation light: ctrl 0.02% ablation 0.02%. For each experiment between 0.02% and 1.7% of bouts were excluded based on those criteria. Data was recorded either at 40Hz (activation experiments) or 160Hz (all other experiments). Effect size was calculated as the difference between the control value and the condition value relative to the control value. For fin body slope effect size the control value of the fast bin (i.e. largest slope) was used for effect size calculations to avoid overestimation of changes due to small control values.

Kinematic analyses proceeded as in [26](#); key parameters were defined as follows:

- **Posture** is the pitch angle of the fish (long axis of the body relative to the horizon) at -250ms relative to peak speed, just before swim bout initiation. Positive values are nose-up.
- **Climb Bouts** are bouts with a trajectory of > 0° at the peak speed of the swim bout.
- **Dive Bouts** are bouts with a trajectory of < 0° at the peak speed of the swim bout.
- **Upward rotation** refers to the rotation from -250ms to the peak angular velocity; only bouts with positive upward rotation were included in the analysis of fin-body coordination.
- **Lift** is the residual change in depth (z) across a bout after subtracting the change expected from the posture of the fish as detailed in [25](#). Briefly, the expected change is calculated using the distance the fish moves in x from -100 to 100ms and the pitch angle at -100ms. Only bouts with positive lift were included in the analysis of fin-body coordination.
- **Fin-lift/rotation coordination** is defined as the slope of the best linear fit between upward rotation and lift across bouts. The goodness of fit, R^2 was used as a measure of how well the fins and trunk are coordinated to generate lift, after [25](#).

Functional GCaMP imaging in Purkinje cells

All calcium imaging experiments were performed using Tilt In Place Microscopy (TIPM), described comprehensively in [68](#). Briefly, 7 dpf fish were mounted in the center of the uncoated side of a mirror galvanometer (catalog #GVS0111, Thorlabs) in 2% low-melting-point agarose. E3 was

placed over the agarose, and the galvanometer mirror was placed under the microscope.

A microscope (Thorlabs Bergamo) was used to measure fluorescence elicited by multiphoton excitation (920nm) from a pulsed infrared laser (Mai Tai HP). Fast volumetric scanning was achieved using a piezo actuator (catalog #PFM450E, Thorlabs) to move the objective. Each frame of the volume (224×96 pixels) was collected with a $0.6 \mu\text{s}$ pixel dwell time (19.1 frames/s) resulting in a sampling rate of 3.82 volumes/s. While this imaging rate might be too slow to distinguish single spikes, it is suitable to measure a difference in calcium transients upon pitch stimulation to nose-up or nose-down direction. To set the galvanometer to a specific angle, a corresponding voltage was applied. The total angular range of the galvanometer is 40° . For the 30° stimuli, the galvanometer was driven to either $+15^\circ$ or -15° and then rotated so that the mirror was horizontal which allowed allowing for a 30° deflection in one direction. For each cell, 21 trials were initially recorded for one stimulus direction. The galvanometer was then remounted to allow for a 30° stimulus in the opposite direction, and 21 trials were similarly recorded for this direction. The order of nose-up and nose-down blocks were alternated for different fish. After all 42 trials were recorded fish were anesthetized with 0.2 mg/ml MESAB; after 10min the baseline fluorescence at $\pm 30^\circ$ was recorded to establish a baseline that controlled for eccentricity. Analysis was done using Fiji and MATLAB. In total 43 Purkinje cells were imaged and 31 cells were kept from 8 fish. Only Purkinje cells that could be reliably identified at $\pm 30^\circ$ were analyzed. To map the anatomical locations of the recorded cells, we imaged overview stacks for each fish. These stacks were manually aligned in Illustrator, and the cells included in the analysis were identified and color-coded according to their tuning properties.

Regions of interest were drawn in Fiji and loaded into MATLAB to extract the intensity of fluorescence after motion correction was performed ¹⁰⁵[\[27\]](#). The integral of each stimulus was calculated and trials of the same direction were averaged as the tonic response to $\pm 30^\circ$ pitch. To extract cells with directional information the directionality index (DI) was calculated by dividing the difference of the up and down responses by the sum of it. Cells with a DI greater than ± 0.35 were considered directionally tuned. Only Purkinje cells that were not directionally tuned were used for principal component analysis and subsequent support vector machine decoding analysis.

To classify trial identity in the dataset, we used a support vector machine (SVM) with a linear kernel. The SVM model was trained using k-fold cross-validation, which splits the data into k subsets (folds). At each iteration, the model was trained on k-1 folds and tested on the remaining fold, ensuring that the model performance was evaluated on unseen data in each fold. Permutations were performed on randomized trial identity as a null hypothesis (5-fold cross-validation; 100 shuffles for randomization). Accuracy was calculated as 1 minus the classification loss.

For calcium imaging in 7 and 14 dpf larvae, a horizontal imaging protocol was used. In total 11 fish were imaged at 7 dpf and 7 fish at 14 dpf. A total of 138/90 (7/14 dpf) cells were recorded. Cells were imaged while the fish was horizontal. For horizontal imaging, we used a $\pm 19^\circ$ stimulus, enabling us to alternate between up and down trials without the need to remount the galvanometer. Before each trial, a 15 sec period was recorded; the average activity during this time was used as the baseline. Fish were pitched nose-down (-19°) for 15s and rapidly returned to horizontal, where-upon calcium activity was measured. This stimulus was then repeated in the nose-up ($+19^\circ$) direction. The maximum dFF of the first second upon return was analyzed. Cells were classified into directional or non-directional based on the directionality index as described above. PCA and decoder analyses were performed using activity from non-directional cells. Decoding accuracy was tested for each fish individually.

Statistics

All statistical testing was done in Matlab R2020a. Data across repeats was pooled for analysis. To assess the variability and determine whether pooling individual experimental repeats within each group was appropriate, we performed a two-way analysis of variance (ANOVA) on the interquartile ranges (IQRs) of the single experimental repeats for the 7 days post-fertilization (dpf) activation, the 7dpf lesion, and the 14dpf lesion experiments without excluding experimental repeats. The results of the ANOVAs and the IQRs for all experimental repeats are reported in **Tables 6** [↗](#) to **11** [↗](#).

To estimate the spread of the data we resampled distributions 100 times with replacement from the data from each condition and computed the expected value for control and perturbed datasets. These permutations were then used to explicitly compute a p-value for fitted variables (slope and R^2 of fin body coordination).

For other variables two-sided Wilcoxon rank sum tests were performed. To correct for multiple testing the critical p-value was calculated based on $\alpha = 0.05$ using Šidák's method. The critical p-value for each data set is reported in the respective table. Outliers were determined as deviating more than 3 times the scaled median absolute deviation (MAD) from the median. A scaling factor of 1.4826 was used to ensure that MAD-based outlier detection is consistent with other methods like Z-scores. Data is shown as median and 95% confidence interval of the median for measured parameters or as median with 25th and 75th percentile for bootstrapped variables. The 95% confidence intervals of the median were bootstrapped using 1000 samples. The medians with 95% confidence intervals for all parameters are reported in the tables. For linear fits a robust regression model (bisquare) was used and fitted variables (slope and R^2 of fin body coordination) were bootstrapped (using 100 samples). To test speed dependency of the fin-lift/rotation ratio Spearman's rank correlation was computed and control and lesion values were compared. First, the correlation coefficients were transformed using Fisher's z-transformation to enable direct comparison of the z-scores. The difference between the z-scores was divided by the standard error and a z-test was performed. Additionally, we only considered effect sizes of $\geq 15\%$ to be biologically relevant.

Data & Code

All data, raw and analyzed, as well as code necessary to generate the figures is available at the following DOI: 10.17605/OSF.IO/9X57Z

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Additional information

Table 5.

Behavior measurements 14 dpf pectoral fin amputation.

	control median [95% CI]	fin amputation median [95% CI]	effect [%]	p-value	significance
fin amputation					
critical p-value: 0.004					
slope slow [mm/s]	0.036 [0.035 – 0.037]	-0.005 [-0.005 – -0.004]	-60	<0.001	yes
slope medium [mm/s]	0.055 [0.054 – 0.056]	-0.005 [-0.005 – -0.005]	-88	<0.001	yes
slope fast [mm/s]	0.068 [0.067 – 0.069]	0.013 [0.013 – 0.013]	-81	<0.001	yes
R ² slow	0.173 [0.162 – 0.187]	0.017 [0.003 – 0.037]	-14	0.141	no
R ² medium	0.370 [0.365 – 0.379]	0.011 [0.003 – 0.019]	-61	<0.001	yes
R ² fast	0.450 [0.441 – 0.458]	0.090 [0.053 – 0.144]	-26	<0.0388	no
Fin lift slow [mm]	0.206 [0.196 – 0.222]	0.066 [0.058 – 0.075]	-68	<0.001	yes
Fin lift medium [mm]	0.219 [0.209 – 0.226]	0.082 [0.078 – 0.087]	-62	<0.001	yes
Fin lift fast [mm]	0.275 [0.263 – 0.284]	0.123 [0.117 – 0.129]	-55	<0.001	yes
Rotation slow [°]	1.544 [1.402 – 1.689]	0.546 [0.455 – 0.652]	-65	<0.001	yes
Rotation medium [°]	1.339 [1.278 – 1.419]	0.642 [0.609 – 0.689]	-52	<0.001	yes
Rotation fast [°]	1.046 [0.975 – 1.119]	0.950 [0.887 – 1.033]	-9	0.014	no

Table 6.

Results of ANOVA on interquartile ranges (IQRs) of single experimental repeats for 7 Days Post-Fertilization (dpf) activation experiments.

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
Group	65.0	3.0	21.7	0.225	0.879
Measurement Type	34.6	4.0	8.7	0.090	0.986
Group*Measurement Type	80.3	12.0	6.7	0.069	1.000
Error	32818.0	340.0	96.5		
Total	32998.0	359.0			

Table 7.

Results of ANOVA on interquartile ranges (IQRs) of single experimental repeats for 7 Days Post-Fertilization (dpf) lesion experiments.

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
Group	15.0	3	5.0	0.064	0.979
Measurement Type	70.0	4	17.5	0.223	0.925
Group*Measurement Type	34.2	12	2.9	0.036	1.000
Error	25069.4	320	78.3		
Total	25188.6	339			

Table 8.

Results of ANOVA on interquartile ranges (IQRs) of single experimental repeats for 14 Days Post-Fertilization (dpf) lesion experiments.

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
Group	8.3	3.0	2.8	0.028	0.994
Measurement Type	382.8	4.0	95.7	0.973	0.423
Group*Measurement Type	47.3	12.0	3.9	0.040	1.000
Error	17700.5	180.0	98.3		
Total	18138.9	199.0			

	climb bout posture [°]	dive bout posture [°]	Duration [s]	IBI [s]	Speed [mm/s]
pre control	20.74	23.65	0.075	1.33	7.36
	28.21	16.96	0.125	1.00	7.28
	29.90	24.49	0.125	1.40	9.43
	24.84	17.06	0.075	1.55	7.96
	22.60	18.18	0.075	1.60	7.64
	23.63	15.56	0.075	2.04	4.99
	18.86	17.54	0.100	2.48	6.69
	22.52	15.44	0.075	2.03	6.93
	23.00	11.93	0.081	2.38	6.92
	20.05	14.85	0.075	1.61	7.30
	19.29	18.75	0.100	2.30	6.00
	22.25	14.81	0.075	2.23	6.25
	21.61	17.06	0.075	1.95	6.36
	24.16	18.94	0.100	1.68	6.66
	19.63	10.03	0.050	1.75	6.68
	22.35	16.96	0.100	1.90	6.44
	21.22	13.89	0.100	2.50	7.08
	21.61	14.43	0.075	1.68	7.10
pre activation	24.68	16.86	0.081	1.87	7.44
	25.48	13.85	0.100	1.30	7.71
	25.88	13.53	0.100	1.08	6.48
	24.92	16.52	0.075	1.68	7.13
	20.30	18.64	0.075	1.38	4.41
	23.32	21.46	0.100	2.99	6.39
	25.03	15.73	0.100	1.83	6.68
	21.49	14.53	0.100	2.12	6.38
	25.84	16.73	0.100	1.58	6.66
	22.33	12.52	0.075	2.10	6.67
	20.44	16.83	0.075	3.10	6.05
	23.67	16.49	0.100	2.78	6.38
	27.07	19.06	0.100	2.70	6.32
	23.81	17.40	0.075	1.66	7.31
	21.82	19.38	0.050	2.33	7.68
	24.27	16.03	0.100	2.33	6.73
	23.44	16.06	0.100	1.98	7.12
	22.89	15.45	0.100	1.58	6.93
control	21.10	16.75	0.050	0.43	5.69
	24.43	43.20	0.100	1.40	7.61
	36.40	18.92	0.100	1.56	8.08
	21.95	18.46	0.075	1.96	7.08
	21.82	20.37	0.075	1.25	8.00
	20.74	19.04	0.075	1.98	6.57
	22.37	20.55	0.075	1.53	6.95
	21.79	13.07	0.075	2.05	7.40
	23.97	16.79	0.075	2.43	6.41
	22.48	16.57	0.075	2.09	6.38
	21.13	21.03	0.075	2.33	6.43
	23.19	17.84	0.075	1.98	6.47
	22.40	18.91	0.075	1.73	6.46
	25.41	20.43	0.075	2.08	6.83
	29.93	18.46	0.075	1.90	5.79
	24.37	25.53	0.100	1.80	6.58
	26.21	19.68	0.100	1.53	6.10
	20.50	18.00	0.075	1.50	6.32
activation	23.40	28.68	0.075	0.65	6.53
	26.09	11.16	0.075	1.65	7.51
	21.35	19.42	0.050	0.75	6.68
	23.69	13.84	0.075	2.31	6.34
	19.90	15.18	0.050	1.25	5.33
	24.64	20.85	0.050	1.96	6.96
	23.64	22.08	0.075	1.78	7.22
	21.78	18.14	0.100	2.19	6.98
	24.24	17.35	0.100	2.57	8.79
	27.93	13.88	0.100	2.63	7.56
	23.48	21.20	0.075	2.51	6.25
	26.16	20.58	0.100	2.33	6.95
	27.49	26.34	0.081	1.73	6.60
	26.88	23.40	0.075	1.85	7.78
	25.26	19.62	0.075	2.10	7.60
	24.41	20.94	0.100	2.33	6.47
	26.64	19.76	0.100	1.90	6.69
	23.86	19.33	0.075	1.53	6.45

Table 9.

IQR for all experimental repeats prior to excluding any repeats for 7dpf Purkinje cell activation data set.

	climb bout posture [°]	dive bout posture [°]	Duration [s]	IBI [s]	Speed [mm/s]
pre control	24.53	14.18	0.113	3.67	9.85
	29.07	16.59	0.088	5.22	6.61
	23.45	16.59	0.094	6.22	7.99
	25.97	15.89	0.125	3.73	9.84
	20.95	15.60	0.094	3.48	6.82
	19.82	19.78	0.100	3.29	9.77
	23.70	13.59	0.119	5.99	7.03
	24.08	18.19	0.095	3.87	8.89
	26.89	12.68	0.088	2.70	6.42
	22.21	14.27	0.113	5.84	7.13
	21.91	17.16	0.113	2.49	9.02
	24.20	14.05	0.131	7.95	6.71
	20.42	16.63	0.088	2.80	5.97
	18.70	11.07	0.094	3.17	6.10
	26.29	10.92	0.094	2.18	6.52
	23.38	11.81	0.063	1.03	7.63
	18.46	13.09	0.106	1.73	7.54
pre lesion	16.27	19.05	0.098	1.64	7.82
	22.59	14.01	0.113	6.83	7.65
	27.26	17.13	0.088	3.37	8.68
	26.93	20.38	0.119	2.15	8.99
	26.21	12.07	0.088	4.38	4.64
	28.53	14.25	0.094	3.81	7.55
	22.40	14.02	0.106	5.10	7.70
	21.12	15.35	0.106	3.70	9.32
	24.34	13.21	0.100	2.75	7.15
	21.10	15.76	0.094	4.42	7.88
	25.81	15.31	0.100	4.41	8.61
	30.09	13.90	0.119	5.43	7.73
	21.85	12.45	0.106	4.18	7.82
	19.56	18.35	0.072	2.10	5.06
	20.57	10.43	0.091	2.39	6.44
	21.79	15.38	0.075	2.69	6.59
	22.69	16.77	0.109	2.09	9.98
post control	22.43	24.44	0.100	2.29	9.60
	21.46	16.25	0.106	6.89	4.19
	29.27	14.18	0.094	3.72	7.25
	22.57	14.49	0.069	3.55	7.13
	20.13	16.04	0.100	4.93	6.48
	23.73	17.81	0.088	3.65	6.99
	25.54	12.29	0.119	4.97	5.39
	16.56	14.54	0.088	3.40	10.50
	25.93	13.75	0.094	4.08	5.71
	24.72	15.81	0.119	6.18	5.81
	23.33	10.90	0.094	2.99	4.77
	23.03	11.87	0.106	7.14	5.50
	20.11	13.99	0.088	3.59	6.06
	18.18	18.27	0.091	2.46	5.63
	21.23	13.98	0.063	0.85	4.65
	20.15	13.44	0.094	1.09	5.63
	20.30	13.38	0.063	0.84	5.27
post lesion	18.49	13.28	0.063	5.34	5.32
	24.95	12.55	0.113	7.19	4.72
	28.40	15.31	0.078	4.29	7.82
	27.02	11.85	0.106	6.30	6.02
	23.87	11.08	0.063	5.39	5.24
	23.90	12.96	0.069	1.96	6.24
	26.79	11.18	0.069	0.91	5.87
	26.27	15.42	0.094	4.43	8.02
	25.66	14.43	0.100	6.43	5.88
	24.62	13.05	0.125	6.24	5.66
	25.76	12.31	0.100	4.98	7.33
	28.85	12.47	0.094	6.11	5.98
	22.35	13.41	0.075	1.85	6.94
	22.36	17.24	0.088	3.17	5.22
	22.52	14.74	0.069	0.88	5.07
	27.76	12.45	0.094	2.01	4.49
	24.21	12.28	0.056	1.16	4.98

Table 10.

IQR for all experimental repeats prior to excluding any repeats for 7dpf Purkinje cell lesion data set.

	climb bout posture [°]	dive bout posture [°]	Duration [s]	IBI [s]	Speed [mm/s]
pre control	28.70	14.05	0.136	5.04	7.83
	26.16	14.40	0.113	4.06	7.27
	35.74	18.05	0.119	2.53	4.47
	30.01	23.57	0.125	1.26	3.32
	30.92	18.92	0.175	3.28	6.19
	29.46	15.84	0.194	3.83	7.89
	6.29	9.94	0.169	1.38	4.81
	15.57	7.98	0.131	3.11	5.13
	20.51	13.12	0.138	4.17	7.72
	20.14	15.36	0.169	3.26	6.63
pre lesion	24.18	13.16	0.131	6.20	5.15
	19.37	18.62	0.138	2.32	7.81
	33.27	18.72	0.122	3.30	5.90
	21.64	14.69	0.128	2.60	5.18
	32.50	16.35	0.169	3.91	6.48
	28.20	17.16	0.169	1.68	6.59
	10.11	6.42	0.150	3.91	4.40
	23.29	17.03	0.100	2.95	5.89
	30.73	10.10	0.150	3.95	8.29
	22.80	14.67	0.156	4.61	8.10
post control	26.02	13.49	0.119	4.14	7.82
	24.71	14.18	0.100	5.50	7.34
	32.91	16.37	0.125	5.63	5.88
	26.24	21.08	0.100	1.75	3.42
	26.15	18.88	0.119	3.98	5.59
	34.05	17.22	0.131	5.06	5.75
	4.98	9.41	0.100	1.74	4.02
	10.19	15.13	0.088	5.12	4.07
	24.15	12.57	0.144	4.92	7.42
	27.81	18.68	0.144	3.74	6.66
post lesion	21.22	13.02	0.106	6.42	4.59
	18.43	11.33	0.106	3.13	7.07
	26.90	16.54	0.119	3.90	5.56
	25.77	20.22	0.094	3.41	4.43
	28.24	16.45	0.113	5.15	5.89
	38.16	16.41	0.106	3.38	4.17
	11.72	7.70	0.091	3.29	3.68
	14.28	8.12	0.119	2.83	4.83
	27.58	11.96	0.125	4.27	5.83
	24.03	16.41	0.113	5.10	7.13

Table 11.

IQR for all experimental repeats prior to excluding any repeats for 14dpf Purkinje cell lesion data set.

Figure S1

Chemogenetic activation of Purkinje cells is reversible.

(A) Calcium imaging time series after 1 μ M Capsaicin of TRPV1-(top) and TRPV1+ cells (bottom) in *Tg(aldoca:TRPV1-tagRFP);Tg(elavl3:h2b-GCaMP6f)* larvae. Two 1.5min time series were recorded showing different cells being active ($dF/F_0 > 2$) at different timepoints (timelapse 1: TRPV1-: 0/16 (0%) cells and TRPV1+: 5/27 (19%) cells activated; timelapse 2: TRPV1-: 0/16 (0%) cells and TRPV1+: 4/27 (15%) cells activated. **(B)** Normalized change in fluorescence following treatment with 1 μ M capsaicin at 6h post treatment and after washout in individual Purkinje cells from *Tg(aldoca:TRPV1-tagRFP);Tg(elavl3:h2b-GCaMP6f)* larvae. **(C)** Example confocal image of Purkinje cell nuclei after 3 and 9h of 1 μ M capsaicin treatment. Speckled fluorescence could be observed after 9h of 1 μ M capsaicin treatment indicative of cell death (white circle). Scale bar 10 μ m.

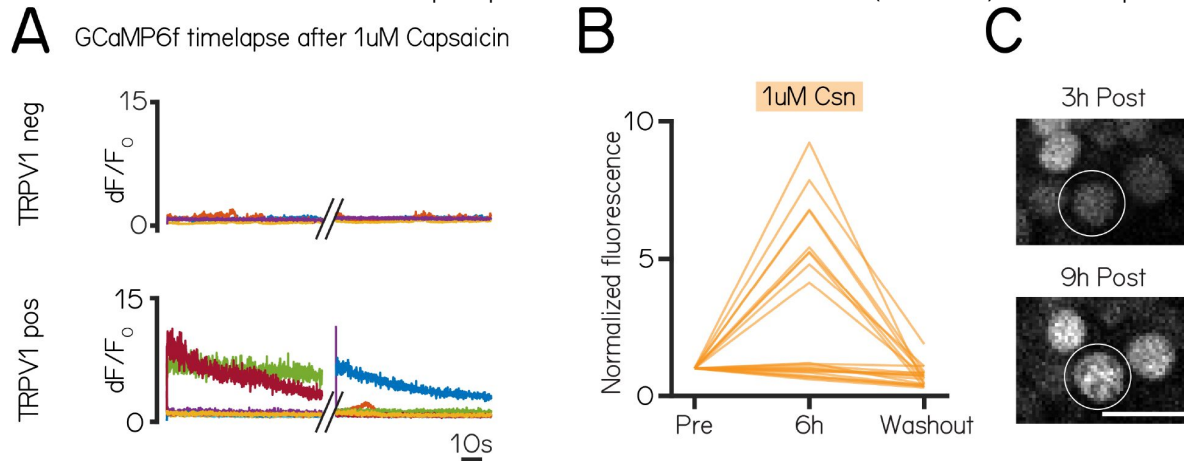


Figure S2

Swim kinematics are not affected by 1 μ M capsaicin treatment.

(A) Distributions of swim kinematics and fin body coordination prior to 1 μ M capsaicin treatment for control group (pre DMSO - grey) and 1 μ M capsaicin group (pre 1 μ M capsaicin - brown) reported in [table 1](#). **(B)** Distributions of swim kinematics and fin body coordination during activation for control (DMSO - grey) and 1 μ M capsaicin treated (orange) groups reported in [table 1](#).

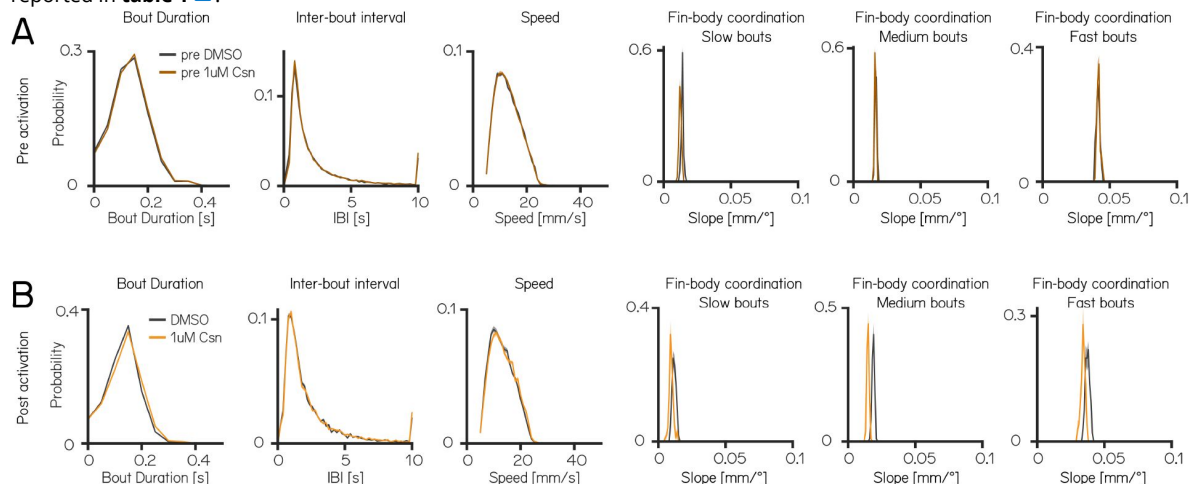


Figure S3

Purkinje cell lesion at 14dpf affects the distribution of postural angles for climb and dive bouts.

(A) Probability distribution of climb postures for control (black) and 10 μ M capsaicin treated 14dpf larvae (red). Data is shown as median and inter-quartile range. (B) Probability distribution of dive postures for control (black) and 10 μ M capsaicin treated 14dpf larvae (red). Data is shown as median and inter-quartile range.

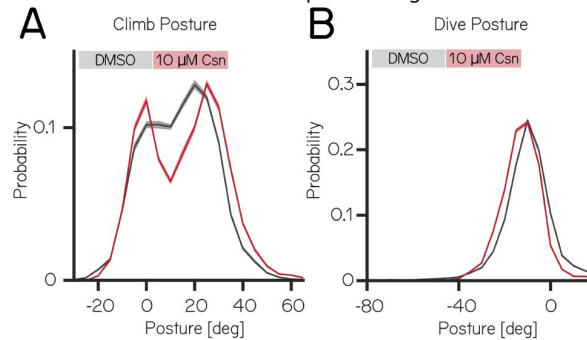
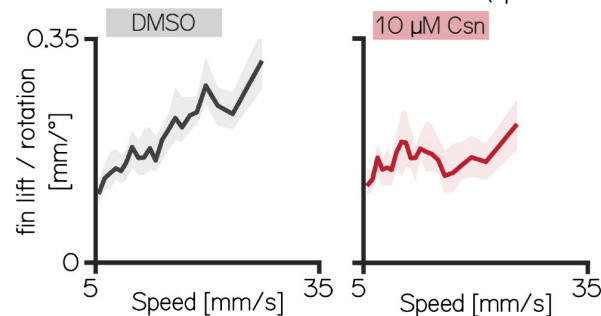


Figure S4

Fin engagement is speed dependent.

(A) Fin Lift / rotation ratio versus speed for 14dpf DMSO treated fish (Spearman correlation coefficient: 0.2193). Data is shown as median with 95% confidence interval of the median. (B) Fin Lift / rotation ratio versus speed for 14dpf fish 10 μ M capsaicin treated fish. Data is shown as median with 95% confidence interval of the median (Spearman correlation coefficient: 0.0397).



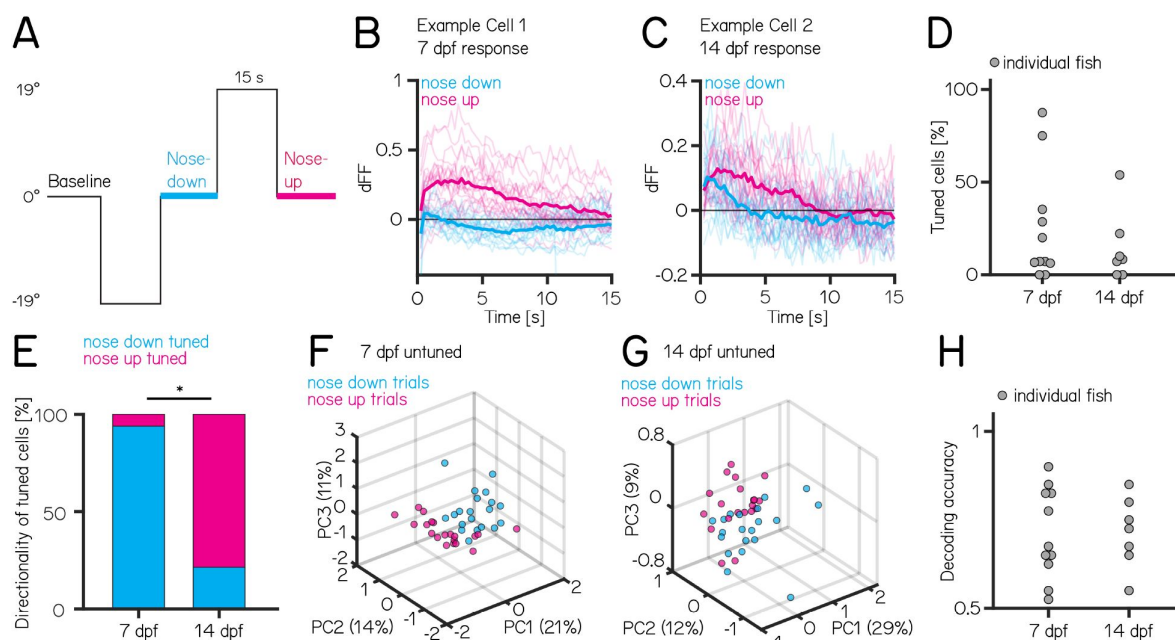


Figure S5

Purkinje cell tuning direction shifts across development, population coding strength remains stable.

(A) One trial consisted of rapid galvanometer steps for 15 seconds in the nose down (-19°, blue) and nose-up (+19°, pink) direction. (B) Example responses (n=40) from a single Purkinje cell at 7 dpf to nose-down (blue) and nose-up (pink) pitch tilts. The thicker lines indicate the median response to all nose-down or nose-up trials. (C) Example responses (n=40) from a single Purkinje cell at 14 dpf to nose-down (blue) and nose-up (pink) pitch tilts. The thicker lines indicate the median response to all nose-down or nose-up trials. (D) Percentage of tuned cells from individual fish based on a directionality index larger than ± 0.35 (median [inter-quartile range]: 7 dpf: 7 [6 – 34]%; 14 dpf: 8 [2 – 19]%; p-value = 0.7763, Wilcoxon rank sum test). (E) Direction of tuned cells at 7 and 14 dpf (7 dpf: 2/31 cells up/down-tuned; 14 dpf: 11/3 cells up/down-tuned, p-value < 0.001, Fisher's exact test). (F) Principal component analysis of all untuned cells at 7 dpf for each of 20 up (pink) and 20 down (blue) trials. (Percentage of variance explained) (G) Principal component analysis of all untuned cells at 14 dpf for each of 20 up (pink) and 20 down (blue) trials. (Percentage of variance explained) (H) Performance of a support vector machine for binary classification of up/down tilt using the responses from untuned neurons. Dots are individual fish at 7 dpf and 14 dpf (median [inter-quartile range]: 7 dpf: 0.68 [0.63 – 0.83]; 14 dpf: 0.73 [0.65 – 0.79]; p-value = 0.9468, Wilcoxon rank sum test).

Author contributions

Conceptualization: FA and DS, Methodology: FA and DS, Investigation: FA and KN, Visualization: FA, Writing: FA, DS Editing: DS, Funding Acquisition: FA and DS, Supervision: DS, MH.
Tg(aldoca:TRPV1-tagRFP) fish were generated by KM.

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Author information

Franziska Auer

Depts. of Otolaryngology, Neuroscience & Physiology, and the Neuroscience Institute, NYU Grossman School of Medicine, New York, United States
ORCID iD: [0000-0002-1193-6002](https://orcid.org/0000-0002-1193-6002)

Katherine Nardone

Depts. of Otolaryngology, Neuroscience & Physiology, and the Neuroscience Institute, NYU Grossman School of Medicine, New York, United States
ORCID iD: [0009-0007-2664-5690](https://orcid.org/0009-0007-2664-5690)

Koji Matsuda

Division of Biological Science, Graduate School of Science, Nagoya University, Nagoya, Japan
ORCID iD: [0009-0008-2537-3627](https://orcid.org/0009-0008-2537-3627)

Masahiko Hibi

Division of Biological Science, Graduate School of Science, Nagoya University, Nagoya, Japan
ORCID iD: [0000-0002-9142-4444](https://orcid.org/0000-0002-9142-4444)

David Schoppik

Depts. of Otolaryngology, Neuroscience & Physiology, and the Neuroscience Institute, NYU
Grossman School of Medicine, New York, United States
ORCID iD: [0000-0001-7969-9632](https://orcid.org/0000-0001-7969-9632)

For correspondence: schoppik@gmail.com

Editors

Reviewing Editor

Filippo Del Bene

Institut de la Vision, Paris, France

Senior Editor

Didier Stainier

Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

Reviewer #1 (Public review):

This study uses a variety of approaches to explore the role of cerebellum, and in particular Purkinje cells (PCs), in the development of postural control in larval zebrafish. A chemogenetic approach is used to either ablate PCs or disrupt their normal activity and a powerful, high-throughput behavioural tracking system then enables quantitative assessment of swim kinematics. Using this strategy, convincing evidence is presented that PCs are required for normal postural control in the pitch axis. Calcium imaging further shows that PCs encode tilt direction. Evidence is also presented that suggests the role of the cerebellum changes over the course of early development, although this claim is less robust. Finally, the authors build on their prior work showing that both axial muscles and pectoral fins contribute to "climbs" and show convincing evidence that PCs are required for speed-dependent engagement of the fins during this behavior. Overall, establishing a role for cerebellum in postural control is not very surprising. However, a clear motivation of this study was to establish a robust experimental platform to investigate the changing role of cerebellar circuits in the development of postural control in the highly experimentally accessible zebrafish larvae and in this regard the authors have certainly succeeded.

This revised version of the manuscript incorporates several improvements. In particular, additional analysis and methodological detail is provided regarding the chemogenetic manipulation, there is expanded analysis of the speed-dependency of pectoral fin engagement, and aspects of the decoding analysis are clearer. However, it is still not certain that the emergence of a dive phenotype over development (from 7 to 14 day post fertilisation) really represents changing role for the cerebellum as opposed to changing sensitivity of Purkinje cells to the chemogenetic treatment.

<https://doi.org/10.7554/eLife.97614.2.sa3>

Reviewer #2 (Public review):

Franziska Auer et al. successfully applied the TRPV1/capsaicin tool to study the contribution of Purkinje cells to postural control. They leveraged the ability of this tool to both activate and ablate neurons within the same construct and tested its effects using their smart, high-throughput behavioral setup for postural control monitoring. With Purkinje cells ablated, balance did not appear to be disrupted; however, postural control was clearly modified along the pitch axis, with larval zebrafish maintaining, on average, a more nose-down posture

compared to controls. While this effect is subtle, it is statistically robust and consistent with the group's previous findings using KillerRed-mediated ablation of Purkinje cells, where the observed postural angle change was explained by a disruption in cerebellar-mediated fin-trunk coordination. Here, the authors present a novel insight, demonstrating that this coordination is swim-speed dependent.

Furthermore, the authors convincingly activated Purkinje cells at 7 dpf, and reported modifications in posture pitch angle comparable to those observed when ablating Purkinje cells. The authors suggest a potential desynchronization of Purkinje cells to explain this observation. Future characterization and application of this activation method to other developmental time points could be of major interest. The authors successfully validated the transfer of the TRPV1/capsaicin method for targeted cell ablation and activation to the study of cerebellar functions and reinforced our current understanding of the role of Purkinje cells in postural control.

This study also explores the developmental evolution of cerebellar function in postural control by comparing the effects of Purkinje cell ablation at 7 dpf and 14 dpf. Interestingly, only dive bout posture showed differential effects across these time points, with no significant impact at 7 dpf but a significant change in postural pitch angle at 14 dpf. In contrast, the effect of Purkinje cell ablation on the climbing bout postural angle remained comparable at both ages. Including additional developmental time points would further strengthen this critical characterization of cerebellar maturation in the context of postural control.

To examine whether Purkinje cell activity encodes postural tilt angle, the authors performed calcium imaging on 31 cells from 8 fish using their Tilt In Place Microscope (TIPM). They found that tilt-angle could be decoded from individual neurons with highly tuned responses, as well as from neurons that were not obviously tuned when pooling their data. The authors refer to this effect as pseudo-population coding because recordings were performed non-simultaneously across animals.

This study successfully integrates cutting-edge genetic tools, high-throughput behavioral assays, and advanced optical microscopy to investigate the role of populations of Purkinje cells in postural control. The authors have not only validated these powerful tools but have also provided novel insights into the cerebellar involvement in postural control, including the swim-speed dependence of fin-trunk coordination.

This work represents an important step toward a detailed understanding of cerebellar contributions to postural control and highlights the potential of combining genetically targeted perturbation with quantitative behavioral analysis.

The authors have addressed my previous concerns, and I congratulate them for their excellent work.

<https://doi.org/10.7554/eLife.97614.2.sa2>

Reviewer #3 (Public review):

Summary:

This paper uses a new chemogenetic tool to investigate the role of cerebellar Purkinje cells in postural control. Using a high-throughput behavioral assay, they show that activation or ablation of Purkinje cells affects various aspects of postural control in zebrafish larvae during spontaneous swimming, and that the effects are more pronounced at later developmental time points, where the Purkinje cell number is much greater. Using a sophisticated imaging assay, they record Purkinje cell activity in response to tilt of the fish, and show that some

Purkinje cells are tuned to tilt direction, and that the direction can even be decoded from untuned neurons.

Strengths:

Overall the study is nice, using a variety of genetic tools and behavioral analysis to address a fundamental question about the role of the cerebellum in postural control in fish

<https://doi.org/10.7554/eLife.97614.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

This study uses a variety of approaches to explore the role of the cerebellum, and in particular Purkinje cells (PCs), in the development of postural control in larval zebrafish. A chemogenetic approach is used to either ablate PCs or disrupt their normal activity and a powerful, high-throughput behavioural tracking system then enables quantitative assessment of swim kinematics. Using this strategy, convincing evidence is presented that PCs are required for normal postural control in the pitch axis. Calcium imaging further shows that PCs encode tilt direction. Evidence is also presented that suggests the role of the cerebellum changes over the course of early development, although this claim is rather less robust in the current version of the paper. Finally, the authors build on their prior work showing that both axial muscles and pectoral fins contribute to "climbs" and show evidence that suggests PCs are required for correct engagement of the fins during this behaviour. Overall, establishing a role for the cerebellum in postural control is not very surprising. However, a clear motivation of this study was to establish a robust experimental platform to investigate the changing role of cerebellar circuits in the development of postural control in the highly experimentally accessible zebrafish larvae, and in this regard, the authors have certainly succeeded.

Overall, I consider this an excellent paper, with some room for improvement in aspects of presentation, discussion, and some aspects of the data analysis..

We thank the reviewer for their kind comments and support. In the revision we have addressed their concerns regarding data presentation and analysis. Additionally, we have expanded our introduction and discussion to address questions of presentation.

Reviewer #2 (Public Review):

Summary:

Franziska Auer et al. investigate the role of cerebellar Purkinje cells in controlling posture in larval zebrafish using the chemogenetic tool TRPV1/capsaicin to bidirectionally manipulate (i.e., activate or ablate) these cells. This tool has been developed for zebrafish previously but has not been applied to Purkinje cells.

High-throughput behavioral experiments are presented to monitor how body posture is affected by these perturbations. The analysis of postural control focuses on a specific subaspect of posture: the body tilt-angle relative to horizontal just before a swim bout is executed, quantified separately for pre-ascent and pre-dive bouts. They report a broad bimodal distribution of pre-ascent bout posture ranging from -20 to +40 degrees, while

the pre-dive bout posture was more Gaussian, ranging between -40 and 0 degrees. The treatment effect is quantified as the change in the median of these distributions.

Purkinje cell activation and ablation in 7 days post-fertilization (dpf) fish shifted the median of the ascending bout posture distributions to positive values. The authors hypothesize that the stochastic nature of the activation process might desynchronize Purkinje cell activity, thus abolishing Purkinje cells' role in postural control, similar to ablation. However, this does not explain why dive bout posture decreased upon activation but was unaffected by ablation.

To test whether the role of Purkinje cells in postural control matures over development, the authors repeated the ablation experiments at 14 dpf. They state that "at 14 dpf, the effects of Purkinje cell lesions on posture were more widespread than at 7 dpf." However, this effect size is comparable to that observed at 7 dpf, suggesting no further maturation of the role of Purkinje cells in pre-ascending bout postural control. The median pre-dive bout posture decreased at 14 dpf, contrasting with no effect at 7 dpf, yet this change was comparable in effect size to the activation effect on Purkinje cells at 7 dpf. The current data breadth may not be sufficient to conclude that signatures of emerging cerebellar control of posture across early development were uncovered.

The study's exploration of activating Purkinje cells in freely swimming fish using TRPV1/capsaicin is of special interest, but the practicability of this method is unclear from the current presentation. It would be beneficial to present the distribution of the percentage of activatable Purkinje cells across animals and time points to provide insight into the method's efficiency. Discussing this limitation and potential improvements would aid in evaluating the method, especially since the authors report that the activation experiments were labor-intensive, limiting repeat experiments. This may explain why the activation experiment at 7 dpf is the only data presented with cell activation, with other analyses performed using the cell ablation capabilities of the TRPV1/capsaicin method.

Another data point at 14dpf would significantly strengthen the conclusions.

The authors analyze Purkinje cell-controlled fin-trunk coordination by examining ascending bout posture across different swim bout speeds. They make the important finding that pectoral fin movements contribute significant lift for median and fast swim bouts but not for slow ones, and that Purkinje cell ablation disrupts lift generation at all speeds.

Finally, the authors examined whether Purkinje cell activity encodes postural tilt-angle by performing calcium imaging on 31 cells from 8 fish using their Tilt In Place Microscope (TIPM). They report that they could decode the tilt-angle from individual neurons with a highly tuned response, and also from neurons that were not obviously tuned when pooling them and analyzing the population response. However, due to the non-simultaneous recordings across animals, definitive conclusions about populationlevel encoding should be made cautiously, it might be better to suggest potential population encoding that needs confirmation with more targeted experiments involving simultaneous recordings.

Strengths:

- The study introduces a novel application of the chemogenetic tool TRPV1/capsaicin to study cerebellar function in zebrafish.
- High-throughput behavioral experiments provide detailed analysis of postural control.
- The further investigation of Purkinje cell-controlled fin-trunk coordination offers new insights into motor control mechanisms.

- The use of calcium imaging to decode postural tilt-angle from Purkinje cell activity presents interesting preliminary results on neuronal population encoding.

Weaknesses:

- The term "disruption" for postural control effects may lead to misleading expectations.
- The supporting data show only subtle median shifts in postural angle, raising questions about the significance of observed effects. Statistical methods that account for the hierarchical structure of the data might be required to support the conclusions.
- The study's data breadth may not be sufficient to conclude emerging cerebellar postural control across early development.
- The current presentation does not adequately detail the practicability and efficiency of the TRPV1/capsaicin method for activating Purkinje cells, and the labor-intensive nature of these experiments constrains the ability to replicate and validate the findings.
- Non-simultaneous recordings in calcium imaging necessitate cautious interpretation of population-level encoding results.

We appreciate the reviewer's thoughtful and detailed feedback. In response, we have made several changes to highlight key points in our manuscript. We have adjusted our wording to more accurately reflect the scope of our findings. Finally, we have clarified and expanded the methods used.

Reviewer #3 (Public Review):

Summary:

This paper uses a new chemogenetic tool to investigate the role of cerebellar Purkinje cells in postural control. Using a high-throughput behavioral assay, they show that activation or ablation of Purkinje cells affects various aspects of postural control in zebrafish larvae during spontaneous swimming and that the effects are more pronounced at later developmental time points, where the Purkinje cell number is much greater. Using a sophisticated imaging assay, they record Purkinje cell activity in response to the tilt of the fish and show that some Purkinje cells are tuned to tilt direction and that the direction can even be decoded from untuned neurons.

Strengths:

Overall the study is nice, using a range of tools to address a fundamental question about the role of the cerebellum in postural control in fish.

Weaknesses:

- (1) *The data in Figure 1 that establishes the method seems to be based on a very small number of experiments and lacks some statistical analysis.*
- (2) *The choice and presentation of the statistical and analysis methods used in Figures 2-5 could be improved.*

We thank the reviewer for their comments. We have added additional statistical analyses for the activation experiments, and improved data presentation .

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Overall I think this is a great paper.

** Introduction and Discussion.*

The Introduction (and Discussion) do little to explain what is understood about cerebellar control of posture and what major outstanding questions remain. The first paragraph of the Introduction seems to argue that the role of the cerebellum in control of posture is well established and line 24 attempts to motivate the present study by virtue of the fact that terrestrial locomotion is "complex". This might be true but is not necessarily a major obstacle given the suite of powerful approaches available in rodent neuroscience. What are the major challenges that are hard to tackle in rodents and what specific questions can the larval zebrafish help to answer? What about development (which gets no mention at all)? I'm not suggesting a comprehensive review of every aspect of cerebellar physiology, but I think the Introduction should attempt to outline the current hypotheses in a little more detail and highlight what we still need to understand.

We take the Reviewer's point that there is more to say in the Introduction. We feel that multi-dimensional limb biomechanics and proprioception are two aspects of terrestrial locomotion that support our use of the word "complexity." However, we don't dwell on this point because, as the reviewer correctly states, the suite of tools for rodent neuroscience & behavior is expansive and, in our opinion, not a limiting factor. Instead, we said what we felt we could regarding the potential contribution of the larval zebrafish in the last paragraph of the Discussion. In the revision, we have added details about the development of cerebellum to the introduction (though this, of course, is an expansive topic and well-beyond the scope of the Introduction), highlighted some of the historical limitations in rodent posture analysis, and set up the .

** Figure 2: 'Arrows denote the shift towards more nose-up postures'. I think the distribution is quite easy to interpret without these arrows; I suggest removing them.*

We have removed the arrows.

** IQR is sometimes stated as a single number and sometimes as a range. It should be consistent and unless eLife has guidance to the contrary, I suggest that it be the latter.*

Thank you for pointing that out. We now report it as the value at the 25&75th %ile for all IQRs.

** Figure S2: For 14 dpf fish the axes are labelled PC2/3 - is this an error?*

We have changed it to a 3-dimensional plot for both 7 and 14 dpf data to show comparable plots for both ages (now Figure S5 F and G). For the analysis in the 14dpf fish the clearest separation was in the space defined by the 2nd and 3rd principal component.

** In the methods, there is insufficient detail given about fluorescent imaging.*

We added additional information to how the fluorescent imaging was performed to the 'Confocal imaging' section as well as to the 'Functional imaging section'

** Abstract*

*In my opinion, the statement "Here, we used a powerful chemogenetic tool (TRPV1/capsaicin) to *define the role of Purkinje cells*..." is too strong. Whilst the evidence that*

PCs are required for postural control is certainly strong, what exactly these cells do in the service of postural control is far from clear (as the authors indeed acknowledge in the Discussion). As such, I wouldn't say their role has been "defined".

We change the word to “describe” to better reflect our findings

** aldoca transgenic.*

This appears to be a beautiful transgenic line but the data showing the extent of its expression and evidence that in the cerebellum it exclusively labels PCs isn't clear enough.

(i) Ideally Figure 1A would show an image of a whole animal to provide an overview of transgene expression but instead it seems to be (the legend is unclear) a cartoon with a confocal projection of part of the brain overlaid.

We have updated the figure legend to be clearer that we show a cartoon of a larval zebrafish with the confocal image overlaid. The aldoca promotor has been previously described and exclusively labels Purkinje cells (10.1523/JNEUROSCI.3352-10.2010)

(ii) Figure 1B shows expression in the cerebellum, but how are we to understand that all the labelled cells are PCs? Are all PCs labelled, or only a subset? Perhaps a double labelling with a PC in situ marker could be done to demonstrate colocalisation?

As above, the aldoca promotor has been previously described; to the best of our knowledge in the Hibi lab's hands (and ours) it labels Purkinje cells exclusively, and it labels all of them (10.1523/JNEUROSCI.3352-10.2010)

** Chemogenetic validation.*

Overall, the chemogenetic approach to abrogate PC function looks to be very powerful. The authors state in several places that a contribution of this paper is in its "establishing the validity of TRPV1/capsaicin-mediated perturbations". However, the data in Figure 1, along with various comments in other parts of the paper raise some questions:

*(i) For experiments depolarising PCs with 1μM CSn, the same size is tiny: Two transgenic animals and one control. Moreover, it is stated 'in one fish ... we observed a small number of neurons at the 9h timepoint with bright, speckled fluorescence suggestive of cell death'. Was this one out of two transgenics?! In the discussion, I didn't understand the statement "ensure adequate brightness levels *to achieve sufficient depolarization without excitotoxicity*". Does this "excitotoxicity" relate to the speckled fluorescence observation?*

Overall, the very small sample size and comments about excitotoxicity and cell death raise concerns about the approach that I think warrant clearer treatment in the results (including information about the assessment of transgene expression, % embryos judged to have suitable expression), especially as this paper is seeking to establish the validity of the method.

We note first that the method has been previously validated (<https://doi.org/10.1038/nmeth.3691>) and that we build on this work. For the experiment described, the point was to identify an acceptable duration for exposure. To that end, we analyzed 6 animals for up to 6h (including the washout experiments in Figure S1B) where we never observed any speckled fluorescence; we limited our behavioral experiments to 6h accordingly. We thought it would be worth including the observation of speckled fluorescence at 9h timepoint for future

reference. To directly address the comment we have increased the number of analyzed cells and fish for the 1uM capsaicin experiments and added statistical analysis (lines 65-67).

When screening for transgene expression we selected for fish that had clearly visible expression, but that did not look overly bright, and used the same criteria when screening fish for the GCaMP imaging and for behavior. Around a quarter of the fish that had aldoca:TRPV1-tagRFP expression had a usable expression level for the activation experiment. We have added this information to the Results (line 62) and Methods (line 369-372)

(ii) The authors note "capsaicin could sporadically activate subsets of Purkinje cells" and further speculate about PC activity and synchrony in the discussion. Figure 1 seems to rely on single images at widely spaced time points but given that they are set up to do 2-photon calcium imaging, why didn't they collect continuous time series data and analyse the temporal patterns of activity across the transgenic PC population?

We have added time series data for calcium imaging after 1uM of Capsaicin in TRPV1- and TRPV1+ cells to Supplementary Figure S1A. Here too we see sporadic increases in calcium levels at similar rates: 0% for TRPV1- and 15-19% for TRPV1+ (see also Figure S1 legend)

(iii) The axonopathy and cell death resulting from 10 μ M Csn is quite dramatic. However, here the authors do not appear to have included a TRPV1 negative control (although oddly they did for 1 μ M treatment) so it is currently unclear whether or not a high conc of Csn alone might be cytotoxic.

Chen et al (<https://doi.org/10.1038/nmeth.3691>) have established the TRPV1/capsaicin method in zebrafish with broad neuronal label and did not see any effect with high doses of capsaicin in TRPV1 negative fish.

** Behavioural assessment - stats*

Overall, the disruption of postural stability after PC manipulations is convincing.

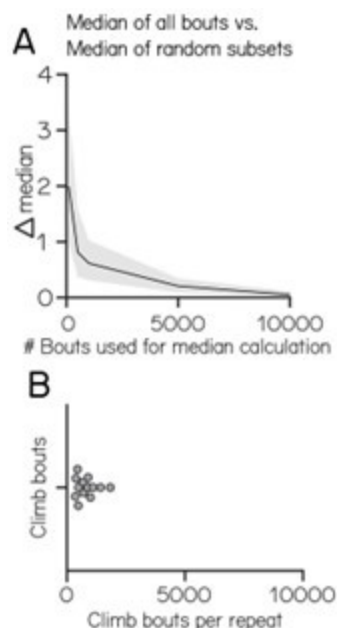
However, I have a few queries about the statistics:

(i) In this section, the statistical unit was not clear. The tables, which are otherwise very useful, give no indication of N. The legend text does report "8 repeats/149 control fish" and "across experimental repeats" suggesting the statistical unit might be the repeats rather than animals, but this should be clarified. In Figure 2G, individual data points should be plotted if N=8, or a representation of the distribution (eg violin or box and whisker plots) if N = 149.

We apologize for the confusion. Given the variable numbers of bouts, a single experimental repeat does not allow for an accurate estimate of expected value. Below we simulated how accurately the median can be estimated based on increasing sample sizes (Author response image 1). Given that large numbers of bouts are necessary to accurately estimate the median we pool the data for all experiments and use resampling statistics to estimate bias in our estimate.

Author response image 1.

Median estimation based on increasing sample size



(ii) Related to the above, I hope it might be easier to interpret the unexpected change in climb posture in ablation controls once the data for individual repeats is shown.

When we analyze the data as single repeats we see considerable variability between different repeats due to undersampling. We tested the medians for the single repeats for outliers to ensure that the shift is not due to a single repeat skewing the distribution. We did not detect any outliers in the pre-lesion control or in the post-lesion control group. (Outliers were determined as deviating more than 3 times the scaled median absolute deviation (MAD) from the median. A scaling factor of 1.4826 was used to ensure that MAD-based outlier detection is consistent with other methods like Z-scores.) We added this information to line 133-134 and the method section under Statistics.

(iii) In some parts of this section, including the Tables, the authors report the 95% CI of the median, rather than IQR. In this case, they should report the z-value used for 95% CI estimation.

As we are using resampling to estimate the 95% confidence interval of the median there is no z-value as in a traditional normal distribution based confidence interval; Instead, we explicitly define the 2.5th and 97.5th percentiles from the bootstrapped sample distribution, which captures the middle 95% of the data, representing the 95% confidence interval.

* It is stated that "fish adopted more nose-up postures before *and throughout* climb bouts". Figure 2F seems to show posture before the climb, but where is the "throughout" data? It would be useful if Figure 2E, J could be extended to make a bit clearer these two phases of postural assessment.

We removed the phrase 'throughout climb bouts' as we are not showing the posture throughout the bout and to avoid over complicating the interpretation.

* Why were PCs not activated at 14 dpf (eg using 1 μ M Csn)?

Due to shifts in priorities the first author will not be continuing this series of experiments, and so this additional experiment will have to wait for someone to pick up this line of inquiry

** The authors appear to claim that the difference in phenotype in 7 versus 14 dpf animals following high conc Csn treatment is indicative of a changing role for cerebellar PCs over this developmental period. For instance, in reference to the 14 dpf ablation phenotype, the authors write "reveals the functional emergence of Purkinje cell control of dives" and in the abstract they talk about "emerging control of posture across early development". However, can they rule out that the phenotypic differences might instead reflect differential sensitivity of the relevant PC (sub)populations to CSn at the two ages? If this caveat cannot be discounted then I suggest it is acknowledged e.g. in the discussion.*

As previously established, all Purkinje cells are labeled in the aldoca line (10.1523/JNEUROSCI.3352-10.2010). Fluorescence is brighter at 14dpf compared to 7dpf, suggesting higher levels of TRPV1. We therefore assume that at 14 dpf, the high concentration of Csn is sufficient to ablate Purkinje cells. At 14 dpf, cerebellar damage is visible under a standard dissecting microscope. The preponderance of evidence therefore speaks against a previously undiscovered subpopulation of TRPV1expressing Purkinje cells that are, by mechanisms yet unknown, resistant to high doses of capsaicin.

** Fin-body "coordination"*

The ideas and data around fin-body coordination are very intriguing.

(i) The statement "fin engagement is speed-dependent" would benefit from a stats test to show this is indeed significant. The data in Figure 4B suggest a rather high degree of variance.

This is an important point; we appreciate the Reviewer's attention. We have added statistics to show this is speed dependent to line 167-169 and show the corresponding plot in the supplement in Figure S4. "Here, we observed that fin engagement is speeddependent, with faster bouts producing greater lift for a given axial rotation (Spearman correlation coefficient: control 0.2193; 10uM capsaicin: 0.0397; Z-test after ztransformation: $p < 0.001$)

*(ii) The statement "After capsaicin exposure, the slopes of the medium fast speed bins were significantly lower (Figure 4C), reflecting *a loss of speed-dependent modulation*" is not convincing. The slope is likely a function of both speed and Csn treatment, and the comparisons in Figure 4C appear to be testing the latter, not the former.*

We understand the reviewer's point. However, the slope for the slow bouts remains unchanged. We therefore conclude that the reduction in fin-body slope is speed dependent and not a speed independent reduction of slope overall.

We have made this more clear by adding Supplementary Figure S4 and changing the text in line 177-179.

(iii) I'd like to understand more about the phenotype of the fin-amputated animals. Were any "bout" parameters changed? Did the animals still attempt climbs and was the distribution of the upward rotation parameter similar to controls? The text states "the slope of the relationship between upward rotation and lift was indistinguishable from zero" but the stats reported in the text are comparisons between groups while Table 5 shows 95% CIs that don't span zero. Some clarification would be useful here.

We appreciate the Reviewer's interest. We've studied climbing in fin-amputated animals at length here: <https://doi.org/10.7554/eLife.45839> and here: <https://doi.org/10.1016/j.celrep.2023.112573> and have added these references in line 183.

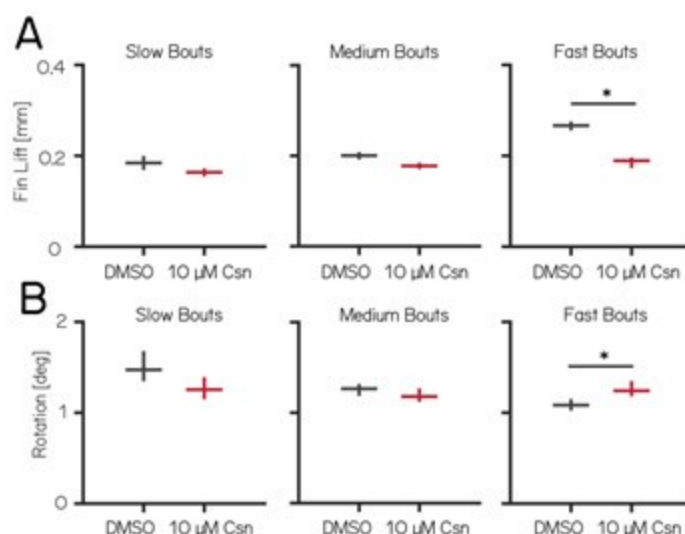
(iv) The authors repeatedly refer to fin-body **coordination** but it is not clear whether the loss of lift after PC ablation is a result of an explicit coordination defect (i.e. changes in the relative timing and/or kinematics between fins and axial motion components), versus a simple reduction in pectoral fin engagement. Either result could be interesting, but this should be clarified.

Thank you for pointing that out. In the fastest speed bin, we observed an increase in upward rotation and a decrease in average fin lift. In contrast, the medium speed bin showed no significant changes in average fin lift or upward rotation (see Author response image 2 and Tables 4 and 5), yet already displayed coordination deficits. Based on these observations, we argue that Purkinje cell lesions primarily affect coordination, rather than simply reducing one specific parameter such as lift or rotation (line 293-298).

We have added fin lift and rotation values from Author response image 2 for all speed bins to tables 4 and 5.

Author response image 2.

Fin lift and rotation for slow, medium and fast bouts



** PC activity and decoding of pitch direction.*

The clever TIPM method is used to collect calcium data that convincingly shows that individual PCs can encode pitch-tilt direction. However, a population of "not tuned" cells are also identified, and here I found the analysis of their responses and the argument that they encode pitch direction at a population level difficult to follow.

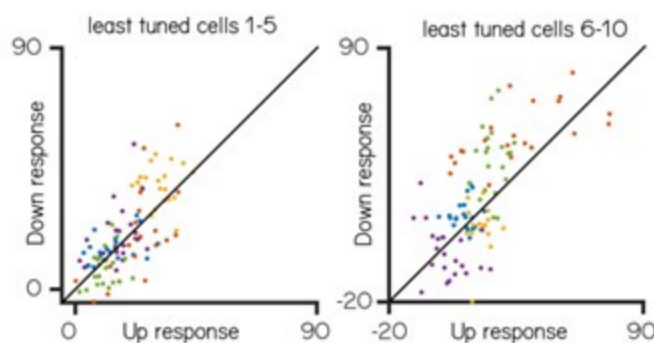
(i) First, although the naming of the cells implies that individual neurons do not encode pitch direction, I did not find this convincing. Figures 5F/G suggest that several "not tuned" cells in fact show quite consistent differences in activity across trial types and indeed in terms of their average responses sit as far from the unity line as do several "tuned" cells.

The Reviewer's comment helped us clarify some key points. First, tuned and untuned cells were categorized based on a Directionality Index threshold of 0.35; some cells might look similar in 5F/G but the highly variable responses of Purkinje cells have highly variable

response so overall there was no consistent tuning. We have clarified this in the text in line 203-207 Below we have plotted the Up versus Down responses for the 10 least tuned cells (sorted by directionality index). While some cells have higher responses on average to one direction we think that the variability makes it difficult to support a claim for “tuning.” We have also tested the support vector machine on the least tuned cells to confirm that the chosen cutoff for tuned/untuned is not affecting our claim that untuned cells can encode position.(see also Author response image 4)

Author response image 3.

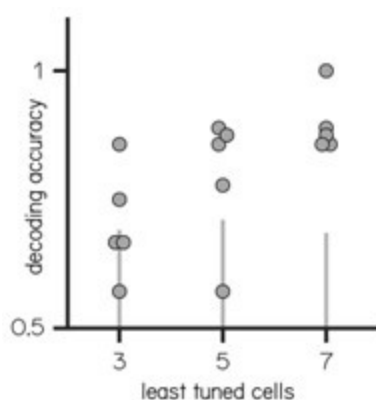
Trial-by-trial variability



(ii) It is therefore not very surprising that PCA (and the SVM decoder) distinguishes trial type. I would guess that PCA assigns the largest weights to these most tuned of the “not tuned” cells, and the 3-5 cell decoders do well when these cells happen to be sampled.

Author response image 4.

Decoding accuracy of the 3/5/7 least tuned cells



This was an interesting idea. To rule out that it is only the most tuned cells that contain the information, we tested the decoder on the 3/5/7 least tuned cells; here too, 5 and more cells are better able to accurately decode the direction. We have add the decoding accuracy to the text in line 221-224

(iii) As I understand the analysis, Figure 5G shows responses for “not tuned” cells over 21 trials (of each type) but these are not the same trials for the different cells? How then is

population coding being assessed?

We have updated the text and refer to this data as a “pseudo-population” in lines 216 and 218 for all experiments where we combined cells from different fish. For technical reasons, when we perform TIPM at eccentric angles we must use sparsely labelled fish to ensure that we can find the same cells over a 60 degree range. We have repeated our analyses for TIPM centered at the horizon, where we can record from entire populations from a single fish.

(iv) Furthermore, Figure S2 shows a somewhat different analysis with decoding accuracy measured on a fish-by-fish basis. In this case, are these decoders for simultaneously imaged neurons? Is this a cross-validated measure of decoding accuracy?

Yes, as above, Figure S4 (former S2) looks at fish-by-fish basis of simultaneous recorded neurons. Yes, it was 5-fold cross validated. We have updated the text in line 490-494.

Reviewer #2 (Recommendations For The Authors):

- Postural control involves various aspects such as balance, coordination, relative body part orientations, and stability. Discussing these and presenting in this context the specific subaspect characterized in this study would help clarify which aspect of postural control the work focuses on.

The Reviewer makes an interesting point, but we think their description of what constitutes postural control is overly broad. Specifically, control of “relative body part orientations in space” by definition requires coordination, and subserves balance and stability. We acknowledge, of course, that different aspects can be and often are treated independently. While interesting, a full treatment of what comprises “postural control” is beyond the scope of the paper, as it would require reconciling the terms across taxa, effectors, environments and well over a century of experiments.

We contend that posture — particularly underwater — is best defined as the relative orientation of body parts in space. For fish, those parts consist of predominantly axial muscles and secondarily fins. We present these definitions in the Introduction and thank the Reviewer for encouraging us to more clearly shape our findings.

- Disruption of posture or postural control: The use of the word "disruption" could lead to misleading expectations. While it may not be incorrect, it suggests a significant loss of equilibrium, an obvious increase in postural variability, or at least a noticeable effect when observing an individual animal's behavior. However, the supporting data show only a subtle median shift in postural angle within a very broad distribution averaged over many individuals. This effect was only significant when comparing fish with a control group, not when comparing fish posture before and after the treatment.

Replacing "disruption" with "modification" would be more cautious.

We take the Reviewer’s point and have adjusted our wording to “modifies postural control.” In lines 137, 266, and 283

*- Statistical significance: Consider aligning the asterisk notation with conventional standards (e.g., * for $p < 0.05$, ** for $p < 0.01$, *** for $p < 0.001$) to enhance clarity for readers. On the other hand, the individual measurements might not be independent (e.g., measurements from the same fish, or the same tank are likely to be correlated), so using the Wilcoxon rank-sum test (Mann-Whitney U test) on pooled data might lead to incorrect conclusions. Methods that account for the hierarchical structure of the data might be required to support the conclusions.*

We take the Reviewer's point about the importance of conventions, however we have never found "more stars = more significant" to be all that helpful in evaluating claims. Instead, we've opted to have both a significance and effect size criteria; a "star" here reflects our considered confidence in the difference we observe.

We agree that the hierarchical nature of pooled data is worth considering/presenting.

We performed a two-way analysis of variance (ANOVA) on the interquartile ranges (IQRs) of the single experimental repeats for the 7 days post-fertilization (dpf) activation, 7dpf lesion, and 14dpf lesion experiments. The ANOVA revealed no significant main effects, supporting the strategy of pooling experimental repeats to estimate distributions.

The results of the ANOVA, along with the IQRs for all experimental repeats, are presented in Tables 6-11. We have also clarified this in the methods section in lines 505-509.

- Data representation: All data of postural angles should be represented in the form of violin plots to show the underlying distributions of the postural angles, especially given that the effect size is small relative to the dispersion of the distribution of the postural angle and that this distribution is also not Gaussian but bimodal, and different before and after the treatments.

We take the Reviewer's point that seeing the full distribution can be useful. We have added plots of the raw distributions for the data in Figure 3 as supplemental Figure S3.

- Showing the distributions will provide the necessary information for the reader to evaluate the importance of the effect. For all data shown in Table 1, the distributions should be presented in the supplementary information.

As requested, we have added the distributions of the data in Table 1 to the supplement (Figure S2)

- Roll posture: A statement about whether roll posture is perturbed by Purkinje cell manipulation would be a piece of important additional information helping to understand how strong the 'disruption' of posture is.

We haven't assessed roll posture, as this is not practical in the current version of the SAMPL apparatus. We have added this limitation to the results (line 116) but also note that as our manipulations are bilateral, we don't anticipate any systematic changes to roll.

- Comparison with other methods: Add a discussion on how the TRPV1/capsaicin method compares with other methods, such as using nitroreductase (Ntr) for targeted pharmacogenetic ablation of cells by treatment with metronidazole or the the possibility to to ablate Purkinje cells by KillerRed as the author lab has done previously. Both methods have been applied to ablate Purkinje cells in larval zebrafish. What are the advantages of the TRPV1 method compared to these when neglecting the activation possibility?

Thank you for that suggestion, we have added a section to the discussion where we compare the TRPV1/capsaicin lesion to other lesion methods (lines 334-336)

- Describe the decoding algorithm: The decoding algorithm used could be described more in detail in the methods section.

We have described the decoding algorithm in more detail in the methods under 'Functional GCaMP imaging in Purkinje cells.' Line 488+

We used a support vector machine (SVM) with a linear kernel. The SVM model was trained using k-fold cross-validation, which splits the data into k subsets (folds). At each iteration, the model was trained on k-1 folds and tested on the remaining fold, ensuring that the model performance was evaluated on unseen data in each fold. Permutations were performed on randomized trial identity as a null hypothesis (5-fold cross-validation; 100 shuffles for randomization). Accuracy was calculated as 1 minus the classification loss.

- Availability of code: The link to the data and code repository is not working.

Thank you for pointing that out, we have fixed it now. In the lower right of the page you can see the history of all changes to the repository, including the entry on 2023-09-08 where the corresponding author set it to “public.” When we checked thanks to your comment, it had been set to “private,” without any record of when/why. We have reset it 2024-10-17. We will continue to check it periodically in the future and apologize in advance if it is unavailable; this is the first time we’ve seen that happen.

- Electrophysiological Control: Including an electrophysiological characterization of the activation of Purkinje cells by the TRPV1/capsaicin would significantly strengthen the validity of the method.

We take the Reviewer’s point that electrophysiological characterization is a way to strengthen the validity of the method. However, Chen et al (<https://doi.org/10.1038/nmeth.3691>) have performed electrophysiology during neuronal activation and concluded that TRPV1 activation with capsaicin indeed increases neuronal activity and firing rates increased. Our calcium imaging and lesion experiments amply demonstrate that Purkinje cells are sensitive to TRPV1-mediated currents. We therefore do not believe that the additional information gained by arduous electrophysiological evaluation is merited here.

- Describe more in detail how climb and dive bouts are defined. The height difference between consecutive bouts measured 250ms before the bout of executions.

Climb and Dive bouts are split by the angle of their trajectory. If the fish moves up (i.e. trajectory larger 0) it is considered a climb bout and vice versa for dive bouts. 250ms prior to the maximum speed is roughly the time the fish initiate a bout, so the pre-bout posture is measured when at this point. The time-courses of bouts are dissected extensively in Zhu et. al. 2023. We have added a definition for climb and dive bouts to the method section under ‘Behavior analysis’ line 453 and 454.

- Figure 1H: Why can't you ablate all Purkinje cells but only about 80%?

This is an excellent question. We opted for an extremely conservative count, and included everything that was still resembling a cell, even if it might not be functional/ already dying. Our counts are therefore likely an underestimate of the percentage of cells that were lost. We have added this point to the text in lines 393 395

• Figure 2C: The method is not fully clear. At 8dpf 0.1uM capsaicin is added to the chamber. At what time after the application of capsaicin did the behavioral recording start?

We recorded after about 10-15min after adding the 1uM Csn to the chambers. The fish were fed after the 6h in capsaicin. We have added this information to the method section line 404 - 408.

- Figure 2F: What indicates the shown confidence interval? Also median with a 95% confidence interval calculated over the experiments in parallel?

The distributions shown in Figure 2F take data from all experiments pooled. We use resampling methods to determine the variability in our estimates. The distribution plots are showing the median and the 25th and 75th percentile of the resampled distribution. We have added this information to the figure legends.

- Figure 3: Subtitles on panel D and E indicating and would facilitate reading.

We have added the subtitles to those panels.

- Figure 4: Describe in the methods how recordings from individual fish were mapped onto each other to superimpose the Purkinje cell locations recorded from the 8 fish.

We have added the respective section to the methods: Line 481 - 483

“To map the anatomical locations of the recorded cells, we imaged overview stacks for each fish. These stacks were manually aligned in Illustrator, and the cells included in the analysis were reidentified and color-coded according to their tuning properties.”

Reviewer #3 (Recommendations For The Authors):

Major points:

(1) Lines 74-81. The data presented here and in later experiments to argue for an effect of capsaicin on neural activity lacks statistical rigor because of the apparently very small numbers of animals/cells assessed. For example, the control appears to involve 4 cells assessed from 1 animal, and the experimental group is just 2 animals. Given that the interpretation of the paper depends upon this result, it is worthwhile to show the result more clearly, and with some statistical analysis. They argue in the discussion that “Our imaging assay established that 1 μ M of capsaicin would stochastically activate subsets of Purkinje cells” which seems a stretch from the data as presented.

We appreciate this point, which was shared by Reviewer 1. We have added more data and performed statistical analysis (line 63 - 67 as well as Figure S1A)

(2) I found the practice of sorting effects by a mixture of effect size and p-value to be a little arbitrary, although in this case, it seems likely that it identified the most relevant effects. I would have preferred to see some attempt to correct for multiple comparisons (e.g. by resampling with the identities of fish shuffled to estimate the distribution of each measurement for this population size), followed by filtering for effect size after establishing a corrected threshold for significance.

We take the Reviewer’s point, though we note that critical values for effect size and pvalue are inevitably “a little arbitrary.” We can’t do the exact analysis the Reviewer suggests as we do not measure data from individual fish for these experiments. However, we did calculate new critical p-values (added to the Tables) that account for multiple comparisons using Šidák’s method.

(3) Figure 4. The data here is a little strange in that the slope in the control condition for medium speed is given as much larger than for slow, but the data in the two cases appears largely overlapping for most of the range of behavior, only diverging for the most extreme rotations. It seems perhaps that the measurement of slope is strongly

dependent on these most extreme values. The authors might want to consider the use of robust regression methods which might mitigate these effects.

This is an interesting observation and we appreciate the Reviewer's thoughtful suggestion. We now use a robust regression method (bisquare weighting of residuals).

We have adjusted all values in lines 175 - 177 and added the regression method to the Methods section line 520.

(4) Figure 5. The 'principal component analysis' description is extremely unclear. The text says that PCA 'showed near-complete segregation of trial types' but it is not explained how this was achieved with PCA or how this was quantified. Figure panels show the data plotted using different pairs of PCs showing visual evidence of segregation. In the methods, it is stated that "We performed principal component analysis" and that "cells were used for principal component analysis and subsequent support vector machine decoding analysis". What is meant exactly by 'performed PCA'? Was PCA used in a dimensionality reduction step? And if so, how many and which PCs were chosen and why? For visualization of the separation, the authors show arbitrary pairs of PCs. Could it be better to use a method more suited to that purpose such as linear discriminant analysis?

PCA was used to define a subspace to qualitatively evaluate if different trials could be separated. Once it became clear that it could, we next trained a binary decoder on the complete dataset (i.e. no dimensionality reduction). We did not perform linear discriminant analysis as the unsupervised PCA already showed separation of trial types. We have made this clearer in lines 212 - 214.

(5) Why does the decoding analysis use only untuned cells? Isn't it equally, or more, interesting to know how well tilt can be encoded using all cells? It is unclear to me what we learn by selecting only untuned cells for this analysis (although I agree it is interesting that this does work).

We focused exclusively on untuned cells because including even a single highly tuned cell for the population coding will lead to excellent results. By using untuned cells we test if there is some directionality information that is not visible just by looking at the up/ down responses of single cells. We have made this clear in lines 217 - 218

Minor points and corrections:

(1) Maybe consider losing the words 'powerful' (I think it is overused and not well defined) and 'reagent'. Reagent is normally used for something that participates in a reaction. It is a bit odd to use it to refer to a transgenic animal. Later it is called a 'tool' which seems better.

We have changed the wording and refer to it as tool for the whole paper.

(2) Figure 1D. Please use a color bar to indicate the scale.

We have added a color scale to the panel

(3) Saying that 'posture' increases is confusing, although the meaning can be inferred from the overall context and the definitions in the Methods - could Posture be capitalized to indicate a specific definition is being used rather than the general meaning?

This suggestion agrees with those made by Reviewer 2. We have changed the wording to “postural angle.”

| *(4) The arrowheads in Figure 2FHK are unnecessary and confusing (why are some horizontal and some vertical?).*

Thank you for that suggestion, we have removed the arrowheads.

| *(5) Figure 3 The legend should indicate that the image is shown with an inverted lookup table.*

We have updated the legend

| *(6) Figure 3 D and E Titles would be helpful, so it is not necessary to refer to the legend to understand the difference.*

We have added titles to the figure panels

| *(7) The dwell time for the 2-photon experiments is given in the manuscript, but I think the authors meant microseconds?*

Thank you for pointing that out. We have corrected it to microseconds.

<https://doi.org/10.7554/eLife.97614.2.sa0>